

AI Usage Declaration

1. General Information

- **Course:** CS202 — Programming Systems (Semester 3 2025–2026)
- **Project:** Plants vs. Zombies (C++20, Raylib, CMake)
- **Group ID:** Group 53
- **Team Members:**
 - 25125086 – Nguyễn Phúc Khánh
 - 25125005 – Từ Hoàng Anh

2. Executive Statement of AI Usage & Academic Integrity

Throughout the development of the **Plants vs. Zombies** remake for the CS202 course, our team utilized the **Google Antigravity** agentic development platform (leveraging Google Gemini 3.6/3.7 Flash and Anthropic Claude Opus 4.6 models).

AI tools served strictly as a collaborative pair-programmer, technical advisor, and rapid prototyping harness for:

1. **Architectural Scaffolding & Design Patterns:** Structuring C++ class hierarchies, planning decoupled game state machines (`MainMenu` , `OptionsMenu` , `ShopMenu` , `SeedSelectMenu` , `Level1` , `BowlingLevel` , `VasebreakerLevel`), and implementing documented OOP design patterns (*Singleton*, *Factory Method*, *Builder*, *Strategy*, *Observer*, *Command*, *State*, *Adapter*, *Facade*).
2. **Animation Reverse-Engineering & Math Modeling:** Decompiling and parsing PopCap Reanim keyframe formats, calculating sub-frame linear interpolation (Lerp) for 60 FPS animation playback, and modeling rolling entity rotational physics, diagonal bounce reflection vectors, and vase shatter physics.
3. **Build Engineering & Cross-Platform CI:** Harmonizing CMake 3.5+ policy shims, resolving Windows MinGW / PowerShell path lockups, automating submission asset downloads (<20MB limit) via GitHub Releases, streamlining Linux / macOS build scripts (`bash build.sh`), and maintaining multi-platform GitHub Actions workflows.

4. **Bitmap Font & Layout Geometry:** Designing parsers for custom PopCap font descriptor files (ContinuumBold , BrianneTod , DwarvenTodcraft , House of Terror , Pico-129) and calculating pixel-perfect button coordinates, conveyor belt slots, and UI clipping bounds.

Verification & Academic Integrity Assertion

In accordance with course guidelines and university academic policies:

- **No unverified code was incorporated:** Every single line of code, math formula, texture coordinate, and architectural design proposed by AI models was manually reviewed, line-by-line evaluated, integrated, and validated locally by the team members.
- **Strict Quality Control:** All changes were tested through local debug builds and validated against automated multi-platform CI/CD pipelines (Ubuntu, macOS, Windows).
- **Transparency:** A full chronological prompt archive, access logs with UTC+7 timestamps, member attributions, and task accomplishment summaries across all 9 primary development streams are documented below.

3. AI Conversation Log

3.1 CMake Automatic Asset Download

- **Conversation ID:** ebe67352-583d-4052-a383-bca0c82b25e7
- **Category / Domain:** Build & Submission Automation
- **Primary Contributor:** Nguyễn Phúc Khánh
- **Date Range (UTC+7):** 2026-09-01 to 2026-09-02
- **Total Active Prompts:** 3

Prompt 3.1.1

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T22:41:44+07:00
- **Prompt Content:**

I need to submit my game project, but the maximum capacity for submission is 20MB. Right now, t

- **Task Accomplished:** Designed CMake custom target and FetchContent/file(DOWNLOAD) script to automatically download assets.zip from GitHub Releases when assets/ is missing, keeping

submission repo size under 20MB.

Prompt 3.1.2

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T22:54:45+07:00
- **Prompt Content:**

/grill-me I've done step 2 using GitHub Releases. The download link is `https://github.com/khanh
What do I do next?
Create a plan.

- **Task Accomplished:** Configured build scripts and submission workflow to automate asset extraction during initial CMake configuration.

Prompt 3.1.3

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-02T12:17:16+07:00
- **Prompt Content:**

/grill-me
Update @[README.md] , in the project installation section:
- `git clone` the `submission` branch
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Designed CMake custom target and FetchContent/file(DOWNLOAD) script to automatically download assets.zip from GitHub Releases when assets/ is missing, keeping submission repo size under 20MB.

3.2 Plant Entity Implementation

- **Conversation ID:** 908043dd-8ecd-4106-85ad-fb49c9f284e5
- **Category / Domain:** Plant Mechanics & Interactions

- **Primary Contributor:** Từ Hoàng Anh
- **Date Range (UTC+7):** 2026-08-31 to 2026-09-01
- **Total Active Prompts:** 18

Prompt 3.2.1

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-31T17:08:24+07:00
- **Prompt Content:**

/grill-me

Fix this bug:

- When a frozen pea passes through the Torch Wood entity, it should revert back to normal pea, I
Create an implementation plan to fix this bug.

- **Task Accomplished:** Fixed projectile interaction where a frozen snow pea passing through Torchwood ignites into a regular flaming pea, resetting slow debuffs and doubling projectile damage.

Prompt 3.2.2

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-31T17:16:57+07:00
- **Prompt Content:**

/grill-me

For the Squash and Ice-shroom plant:

- Find their respective sound effects in @[assets] , and play them when they trigger their effect
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Implemented specialized plant entity combat behavior, animation state transitions, and projectile spawning.

Prompt 3.2.3

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-31T17:25:11+07:00

- **Prompt Content:**

/graphify Can the fog in night levels be removed one lawn tile at a time? That is, is the fog in

- **Task Accomplished:** Implemented specialized plant entity combat behavior, animation state transitions, and projectile spawning.

Prompt 3.2.4

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-31T17:32:47+07:00
- **Prompt Content:**

/grill-me

Create a new Lantern plant:

- Find in @[assets] all assets related to the lantern plant, based on the other implemented plants.
- Add the plant to the Plant Almanac, with its own description, and animation.

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Implemented specialized plant entity combat behavior, animation state transitions, and projectile spawning.

Prompt 3.2.5

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-31T17:52:35+07:00
- **Prompt Content:**

/grill-me

For the Plantern plant:

- In a night level, when a Plantern plant is planted: The fog on the plant's row, one row above.
- If multiple Planterns are planted, the fog will move to the possible place farthest to the right.
- When a Plantern is destroyed (shoveled or eaten by zombies): The fog moves back to its correct position.

Create an implementation plan, and wait for my approval.

- **Task Accomplished:** Implemented specialized plant entity combat behavior, animation state transitions, and projectile spawning.

Prompt 3.2.6

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-31T18:04:38+07:00
- **Prompt Content:**

/grill-me /goal

In the Plant Almanac, change the lawn background of Plantern to night lawn.

- **Task Accomplished:** Implemented specialized plant entity combat behavior, animation state transitions, and projectile spawning.

Prompt 3.2.7

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-31T21:15:28+07:00
- **Prompt Content:**

/grill-me

Fix this bug: In level 1, I used a CherryBomb plant, but it killed a zombie 2 rows below it. Your task is: Check and make sure the CherryBomb plant and the Explode-o-nut only kills zombies Find the root cause of the bug, and create an implementation plan to fix it.

- **Task Accomplished:** Implemented specialized plant entity combat behavior, animation state transitions, and projectile spawning.

Prompt 3.2.8

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-31T21:29:37+07:00
- **Prompt Content:**

/grill-me

You changed the range too. Revert the range back, I just want the explicit row check.

- **Task Accomplished:** Implemented specialized plant entity combat behavior, animation state transitions, and projectile spawning.

Prompt 3.2.9

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-31T21:35:04+07:00
- **Prompt Content:**

/grill-me Is everything else the same, and the only difference is the explicit row checking?

- **Task Accomplished:** Implemented specialized plant entity combat behavior, animation state transitions, and projectile spawning.

Prompt 3.2.10

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-31T22:20:55+07:00
- **Prompt Content:**

/graphify

Does the Squash plant kill all zombies on an adjacent lawn tile?

- **Task Accomplished:** Implemented specialized plant entity combat behavior, animation state transitions, and projectile spawning.

Prompt 3.2.11

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-31T22:31:52+07:00
- **Prompt Content:**

/graphify Does the pole-vaulting zombie jump over the spikerock plant or the spikeweed plant?

- **Task Accomplished:** Integrated multi-hit damage absorption and continuous ground puncture mechanics for Spikerock and Spikeweed entities.

Prompt 3.2.12

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-31T22:34:14+07:00
- **Prompt Content:**

/grill-me

Create a new Twin Sunflower plant:

- Find in @[assets] all assets related to the Twin Sunflower plant, based on the other implement
- Add the plant to the Plant Almanac, with its own description, and animation.

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Implemented specialized plant entity combat behavior, animation state transitions, and projectile spawning.

Prompt 3.2.13

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-31T22:39:33+07:00
- **Prompt Content:**

/grill-me

Update the implementation plan:

- Change the cost of Twin Sunflower to 125.
- Also, the Twin Sunflower sprite doesn't look right. Inspect the `reanim` file again and fix it

- **Task Accomplished:** Implemented specialized plant entity combat behavior, animation state transitions, and projectile spawning.

Prompt 3.2.14

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-09-01T16:57:08+07:00
- **Prompt Content:**

/grill-me

There is a bug: When the zombie bites the garlic in a level, the zombie doesn't play the `yuck`
Create an implementation plan, and wait for my approval.

- **Task Accomplished:** Implemented specialized plant entity combat behavior, animation state transitions, and projectile spawning.

Prompt 3.2.15

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-09-01T17:06:19+07:00
- **Prompt Content:**

/grill-me

There is a bug: When the chomper bites the zombie in a level, the chomper doesn't play the `chor`
Create an implementation plan, and wait for my approval.

- **Task Accomplished:** Implemented Chomper bite detection, swallowing state machine, and chew cooldown timer before returning to idle.

Prompt 3.2.16

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-09-01T17:36:28+07:00
- **Prompt Content:**

/graphify What is the waiting time for the Potato Mine to grow? Also, search in the @[assets] fo

- **Task Accomplished:** Implemented specialized plant entity combat behavior, animation state transitions, and projectile spawning.

Prompt 3.2.17

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-09-01T17:41:25+07:00
- **Prompt Content:**

/grill-me

Fix the PotatoMine and its explosion:

- Change the delay time from 3 secs to 15 secs.
 - From the list of Assets Related to the Potato Mine "SPUDOW" Explosion above, implement the mission.
- Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Implemented specialized plant entity combat behavior, animation state transitions, and projectile spawning.

Prompt 3.2.18

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-09-01T22:43:37+07:00
- **Prompt Content:**

/graphify What do the cabbagepult and cornpult do?

- **Task Accomplished:** Analyzed lobbed-pult projectile trajectories and firing cooldown timers for Cabbage-pult and Corn-pult entities.

3.3 Bowling Level Implementation

- **Conversation ID:** 489f69e9-62d0-4af3-8e80-987445d02f53
- **Category / Domain:** Wall-nut Bowling Level
- **Primary Contributor:** Nguyễn Phúc Khánh
- **Date Range (UTC+7):** 2026-07-30 to 2026-09-01
- **Total Active Prompts:** 82

Prompt 3.3.1

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-30T22:52:28+07:00
- **Prompt Content:**

/grill-me

You MUST READ AND ANALYZE ALL OF THE UPDATED CODEBASE in `CS202-PlantsVsZombies`. Spawn subagent
Now, the tasks are:

- Create a `Zen Garden` button in the Main Menu screen. Use the `Zen Garden` assets in @[assets]
- Clicking the `Zen Garden` button brings me to the `Testing` menu, and clicking `Start Adventure`
- Use pixel-perfect hover checking for that button.

Create an implementation plan first for these tasks, and wait for my approval.

- **Task Accomplished:** Integrated wave spawner manager in BowlingLevel.cpp , scheduling zombie waves across 5 lanes with multi-hit armor damage states.

Prompt 3.3.2

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-30T22:58:33+07:00
- **Prompt Content:**

/grill-me /goal

Tiny formatting task: Move the `Zen Garden` button to `x=300, y=425`.

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.3

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-30T23:01:01+07:00
- **Prompt Content:**

/grill-me /goal

YOU MUST READ, ANALYZE, AND UNDERSTAND ALL FILES OF THE UPDATED CODEBASE in `CS202-PlantsVsZomb:

- **Task Accomplished:** Integrated wave spawner manager in BowlingLevel.cpp , scheduling zombie waves across 5 lanes with multi-hit armor damage states.

Prompt 3.3.4

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-01T11:48:55+07:00
- **Prompt Content:**

/grill-me

Now I want the "Start Adventure" button to bring me to the "Wall-nut bowling" level in PvZ.

Next tasks:

- In the Bowling level, just draw the front lawn image at daytime.
- When I hover on a square in the front lawn, it should draw a rectangle of that square, just 1: Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.5

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-01T12:49:41+07:00
- **Prompt Content:**

One question: Did you do the hover rectangle thing exactly IDENTICAL the "Level 1"?

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.6

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-01T12:52:08+07:00
- **Prompt Content:**

/grill-me /goal

Next task:

- Draw the @[assets/images/Wallnut_bowlingstripe.png] right between the 3rd and 4th column of the

- **Task Accomplished:** Implemented card pick-and-place cursor logic, restricting valid plant placement tiles strictly to columns 1–3 behind the red bowling line.

Prompt 3.3.7

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-02T11:10:49+07:00
- **Prompt Content:**

/grill-me

Now, next tasks:

- Draw the @[assets/images/ConveyorBelt_backdrop.png] in Bowling level exactly where the seed box is
- In @[assets/images/ConveyorBelt.png] , there are 6 rows of conveyor belt textures to animate 1
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Engineered conveyor belt UI with 6-row animated sprite sheet textures, automated 3s card generation, and smooth slot sliding physics.

Prompt 3.3.8

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-02T11:16:41+07:00
- **Prompt Content:**

/grill-me /goal

Tiny fix:

- You MUST analyze @[assets/images/ConveyorBelt_backdrop.png] .
- The backdrop image has a steel frame. Move the animating @[assets/images/ConveyorBelt.png] up

- **Task Accomplished:** Engineered conveyor belt UI with 6-row animated sprite sheet textures, automated 3s card generation, and smooth slot sliding physics.

Prompt 3.3.9

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-02T17:20:11+07:00
- **Prompt Content:**

/grill-me

Next tasks:

- For every 3 seconds, spawn a Wallnut card in the conveyor belt in the Bowling level.
- Each card must spawn on the right of the conveyor belt and slowly moves to the left of the cor
- The horizontal spacings between the cards are 0.

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Engineered conveyor belt UI with 6-row animated sprite sheet textures, automated 3s card generation, and smooth slot sliding physics.

Prompt 3.3.10

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-02T17:26:10+07:00
- **Prompt Content:**

Continue with the implementation plan.

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.11

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-02T17:38:53+07:00
- **Prompt Content:**

/grill-me /goal

Bug fix: The positions of the walnut cards are not right, like in the image.

Find the root cause of the bug and fix it.

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.12

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-02T17:43:45+07:00
- **Prompt Content:**

/grill-me /goal

Bug fix: The positions of the walnut cards in the image don't fill the entire conveyor belt.

Find the root cause of the bug and fix it.

- **Task Accomplished:** Engineered conveyor belt UI with 6-row animated sprite sheet textures, automated 3s card generation, and smooth slot sliding physics.

Prompt 3.3.13

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-02T17:51:22+07:00
- **Prompt Content:**

/learn and also read the rules in @[.agents/rules/agents.md] , @[.agents/rules/graphify.md] .

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.14

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-02T17:54:20+07:00
- **Prompt Content:**

Add and specify that it is for the wall-nut bowling level.

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.15

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-02T21:04:55+07:00
- **Prompt Content:**

/grill-me

Next feature:

- I want to click to select a Wall-nut card in the conveyor belt, and place them on the lawn fire
 - The conveyor belt will refill until full of cards.
 - Write the code to be similar to the "Level 1" code in @[include/level/Level1.h] and @[src/lev
- Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Engineered conveyor belt UI with 6-row animated sprite sheet textures, automated 3s card generation, and smooth slot sliding physics.

Prompt 3.3.16

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-02T21:14:12+07:00
- **Prompt Content:**

/grill-me

Minor change: DO NOT ALLOW deselecting a card. When clicking to select a card, the card is removed.
Update the implementation plan.

- **Task Accomplished:** Engineered conveyor belt UI with 6-row animated sprite sheet textures, automated 3s card generation, and smooth slot sliding physics.

Prompt 3.3.17

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-02T21:44:34+07:00
- **Prompt Content:**

/grill-me

Next feature:

- When I place down a Wall-nut card, instead of staying on the tile indefinitely, it should roll.
 - The tile is immediately available after the Wall-nut rolls for other Wall-nuts to be placed on it.
- Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Implemented card pick-and-place cursor logic, restricting valid plant placement tiles strictly to columns 1–3 behind the red bowling line.

Prompt 3.3.18

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-02T21:55:23+07:00
- **Prompt Content:**

/grill-me /goal

Minor fix: In the Bowling level, when I click on a Wall-nut card and place it on the field, it's not

- **Task Accomplished:** Implemented card pick-and-place cursor logic, restricting valid plant placement tiles strictly to columns 1–3 behind the red bowling line.

Prompt 3.3.19

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-03T16:41:57+07:00
- **Prompt Content:**

/grill-me

Next features:

- Like Level 1, right-click on a tile to spawn a zombie entity on the right of that lane.
 - Like Level 1, have the zombies walk from the right to the left. Display the same "Game over" ;
- Code these 2 features identical to Level 1's code in @[include/level/Level1.h] and @[src/level/]
- Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Integrated wave spawner manager in BowlingLevel.cpp , scheduling zombie waves across 5 lanes with multi-hit armor damage states.

Prompt 3.3.20

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-03T21:16:52+07:00
- **Prompt Content:**

/grill-me

Next feature:

- Collision detection: Check whether a rolling Wall-nut collides with a zombie (use the distance to the zombie's position)
- If collision occurs, the Wall-nut changes rolling direction in the y-axis (the Wall-nut still rolls horizontally)
- If rolling completely horizontally, the velocity changes to add upward or downward y-axis component
 - If rolling with upward/downward y-axis component, y-axis component gets multiplied by -1 (flip direction)
- Don't change the zombie's HP for now.
- Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Engineered circle-based collision detection, calculating diagonal deflection reflection vectors on zombie hit and bouncing off lawn top/bottom boundaries.

Prompt 3.3.21

- **Member:** Nguyễn Phúc Khánh

- **Sent At:** 2026-08-03T21:43:08+07:00
- **Prompt Content:**

/grill-me /goal

Minor changes:

- Condition for "Lawn Boundary Bounce": when center of Wall-nut hits the top edge of the tiles c

- **Task Accomplished:** Engineered circle-based collision detection, calculating diagonal deflection reflection vectors on zombie hit and bouncing off lawn top/bottom boundaries.

Prompt 3.3.22

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-03T21:54:18+07:00
- **Prompt Content:**

/grill-me

There is a bug:

When a wallnut hits a zombies, it changes velocity once, but then changes velocity for a second Find the root cause of this bug, and fix it.

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Engineered circle-based collision detection, calculating diagonal deflection reflection vectors on zombie hit and bouncing off lawn top/bottom boundaries.

Prompt 3.3.23

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-03T22:13:15+07:00
- **Prompt Content:**

/grill-me

There is a bug:

When there are 2 zombies on the same lane and close to each other, then the incoming wall-nut h: Find the root cause of this bug, and fix it.

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Engineered circle-based collision detection, calculating diagonal deflection reflection vectors on zombie hit and bouncing off lawn top/bottom boundaries.

Prompt 3.3.24

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-03T22:28:39+07:00
- **Prompt Content:**

/grill-me

There is a bug:

When a Wall-nut goes behind a zombie but is still in the zombie's range, it still counts as collision. Find the root cause of this bug, and fix it.

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Engineered circle-based collision detection, calculating diagonal deflection reflection vectors on zombie hit and bouncing off lawn top/bottom boundaries.

Prompt 3.3.25

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-04T21:56:59+07:00
- **Prompt Content:**

/grill-me

Next tasks, for debug purposes:

- Draw red bounding boxes and the center points of all Wall-nuts and Zombies.
 - If a Wall-nut collides with a zombie, based on collision detection, the bounding box and center point of the zombie.
 - From the method you implemented for the collision detection, draw a semi-transparent area for the zombie's range.
- Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Implemented card pick-and-place cursor logic, restricting valid plant placement tiles strictly to columns 1–3 behind the red bowling line.

Prompt 3.3.26

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-04T22:20:26+07:00
- **Prompt Content:**

/grill-me /goal

Minor change: In the bowling level, add a simple toggle button called "Debug" to toggle display:

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.27

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-04T22:38:00+07:00
- **Prompt Content:**

/grill-me

Minor change:

- The y-coordinates of the centers of a zombie and a Wall-nut in the same lane are different. No
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Integrated wave spawner manager in `BowlingLevel.cpp`, scheduling zombie waves across 5 lanes with multi-hit armor damage states.

Prompt 3.3.28

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-04T23:19:43+07:00
- **Prompt Content:**

/grill-me

Clarification: Revert the center back to where it was.

What I meant was that the zombie's center for collision detection should be level with the Wall-nut.
Create an implementation plan first to fix the bug, and wait for my approval.

- **Task Accomplished:** Engineered circle-based collision detection, calculating diagonal deflection reflection vectors on zombie hit and bouncing off lawn top/bottom boundaries.

Prompt 3.3.29

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-05T20:44:15+07:00
- **Prompt Content:**

/graphify

Find me all asset files related to the "Giant Wall-nut" and the "Explode-o-nut" in the Wall-nut

- **Task Accomplished:** Created Giant Wall-nut entity with heavy non-deflecting rolling mechanics that plow straight through zombies in its lane.

Prompt 3.3.30

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-05T20:56:23+07:00
- **Prompt Content:**

/grill-me

Now, based on the information above, do this next feature:

- Create a Giant Wall-nut seed packet, it can spawn in the conveyor belt
 - I can place Giant Wall-nut entities. But Giant Wall-nut entities don't bounce, they keep moving
- Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Engineered conveyor belt UI with 6-row animated sprite sheet textures, automated 3s card generation, and smooth slot sliding physics.

Prompt 3.3.31

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-05T21:26:03+07:00
- **Prompt Content:**

/grill-me

Edit the plan: Only do collision detection of Giant Wall-nut across its own lane, IGNORE ALL OTHERS

- **Task Accomplished:** Created Giant Wall-nut entity with heavy non-deflecting rolling mechanics that plow straight through zombies in its lane.

Prompt 3.3.32

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-05T21:46:14+07:00
- **Prompt Content:**

/graphify Check if the Giant Wall-nut gets deleted properly when leaving the screen exactly like

- **Task Accomplished:** Created Giant Wall-nut entity with heavy non-deflecting rolling mechanics that plow straight through zombies in its lane.

Prompt 3.3.33

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-06T21:27:29+07:00
- **Prompt Content:**

/grill-me

Now, based on the information above, do this next feature:

- Create an "Explode-o-nut" seed packet, it can spawn in the conveyor belt.
 - I can place Explode-o-nut entities. But for now, Explode-o-nut entities don't bounce, they keep moving straight.
- Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Engineered conveyor belt UI with 6-row animated sprite sheet textures, automated 3s card generation, and smooth slot sliding physics.

Prompt 3.3.34

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-06T22:47:33+07:00
- **Prompt Content:**

/grill-me

Now refactor the code:

- Use different classes for the normal Bowling Wall-nut, Giant Wall-nut, and Explode-o-nut.

YOU MUST KEEP ALL FUNCTIONALITIES THE SAME.

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Created Giant Wall-nut entity with heavy non-deflecting rolling mechanics that plow straight through zombies in its lane.

Prompt 3.3.35

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-06T23:12:06+07:00
- **Prompt Content:**

/grill-me /goal

Minor change:

- For the seed packet textures of Giant Wall-nut and Explode-o-nut, use @[assets/PlantSeedPacket

- Remove the red tint of the Explode-o-nut seed packet card.

No need to create an implementation plan, proceed immediately.

- **Task Accomplished:** Implemented card pick-and-place cursor logic, restricting valid plant placement tiles strictly to columns 1–3 behind the red bowling line.

Prompt 3.3.36

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-07T10:35:58+07:00
- **Prompt Content:**

/graphify

How is spawning zombies implemented in the Level 1?

- **Task Accomplished:** Integrated wave spawner manager in BowlingLevel.cpp , scheduling zombie waves across 5 lanes with multi-hit armor damage states.

Prompt 3.3.37

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-07T10:37:51+07:00
- **Prompt Content:**

/grill-me

Now, next feature for Bowling level:

- From the information above, create the spawning zombie mechanics similar to that of "Level 1".
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Integrated wave spawner manager in BowlingLevel.cpp , scheduling zombie waves across 5 lanes with multi-hit armor damage states.

Prompt 3.3.38

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-07T13:16:01+07:00
- **Prompt Content:**

/grill-me

Next feature:

In the Bowling level, when a normal Wall-nut collides with a zombie, deal damage based on the zc
- Normal Zombie/Flag Zombie: Dies.
- Conehead Zombie: One hit -> loses the cone, 2 hits -> dies.
- Buckethead Zombie: 1 hit -> heavily dented bucket, 2 hits -> loses the bucket, 3 hits -> dies.
Ignore the Giant Wall-nut and Explode-o-nut for now.
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Created Giant Wall-nut entity with heavy non-deflecting rolling mechanics that plow straight through zombies in its lane.

Prompt 3.3.39

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-07T14:03:21+07:00
- **Prompt Content:**

/grill-me

Bug fix: Add the zombie death animation, I don't see it during the gameplay.

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Integrated wave spawner manager in `BowlingLevel.cpp`, scheduling zombie waves across 5 lanes with multi-hit armor damage states.

Prompt 3.3.40

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-07T21:46:02+07:00
- **Prompt Content:**

/grill-me

Next feature:

- In the Bowling level, when a Giant Wall-nut collides with a zombie, the zombie immediately dies.
 - All zombie death animations must be implemented properly.
 - The Conehead and Buckethead Zombies also play the falling cone and bucket animations, respectively.
- Ignore the Explode-o-nut for now.

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Created Giant Wall-nut entity with heavy non-deflecting rolling mechanics that plow straight through zombies in its lane.

Prompt 3.3.41

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-07T22:14:20+07:00
- **Prompt Content:**

/graphify

What is the range of explosion of the `CherryBomb` where the zombie will be instantly killed?

- **Task Accomplished:** Integrated wave spawner manager in `BowlingLevel.cpp`, scheduling zombie waves across 5 lanes with multi-hit armor damage states.

Prompt 3.3.42

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-07T22:21:42+07:00
- **Prompt Content:**

/grill-me

Using the information above, do this next feature:

- In the Bowling level, when an Explode-o-nut collides with a zombie, it immediately explodes 1:
- Implement the explosion range and damage mechanism exactly like the `CherryBomb`.
- All zombie death animations must be implemented properly, with ash animations.

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Implemented Explode-o-Nut entity with 3x3 lawn grid explosive area-of-effect damage, charred ash animation, and explosion particles.

Prompt 3.3.43

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-07T22:38:36+07:00
- **Prompt Content:**

/grill-me

Fix this bug: I don't see the explosion animations like the `CherryBomb` in Level 1, I only saw
Implement the explosion animations similar to the `CherryBomb` in Level 1.
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Implemented card pick-and-place cursor logic, restricting valid plant placement tiles strictly to columns 1–3 behind the red bowling line.

Prompt 3.3.44

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-08T12:59:18+07:00
- **Prompt Content:**

/grill-me /goal

Fix this minor visual bug:

- Move the center of drawing (displaying) the Giant Wall-nut upward by 20px. You must keep the c

- **Task Accomplished:** Created Giant Wall-nut entity with heavy non-deflecting rolling mechanics that plow straight through zombies in its lane.

Prompt 3.3.45

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-08T21:38:14+07:00
- **Prompt Content:**

/grill-me

Next task:

- Explain what the merge conflicts are, and fix all the merge conflicts.

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.46

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-11T21:13:28+07:00
- **Prompt Content:**

/grill-me /goal

Next task:

- Update the damage of Explode-o-nut to 1800

- **Task Accomplished:** Implemented Explode-o-Nut entity with 3x3 lawn grid explosive area-of-effect damage, charred ash animation, and explosion particles.

Prompt 3.3.47

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-11T21:41:35+07:00
- **Prompt Content:**

/graphify

The zombie and wall-nut entities draw a red dot as collision center when debug mode is on. Check

- **Task Accomplished:** Implemented card pick-and-place cursor logic, restricting valid plant placement tiles strictly to columns 1–3 behind the red bowling line.

Prompt 3.3.48

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-11T21:46:08+07:00
- **Prompt Content:**

/graphify

Check if the collision center coordinates of BOTH the zombies and wall-nuts are centered on the

- **Task Accomplished:** Implemented card pick-and-place cursor logic, restricting valid plant placement tiles strictly to columns 1–3 behind the red bowling line.

Prompt 3.3.49

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-11T23:01:06+07:00
- **Prompt Content:**

/grill-me /goal

Next task:

- Update the range of explosion of the Explode-o-nut (the range where if a zombie's center of c

- **Task Accomplished:** Implemented Explode-o-Nut entity with 3x3 lawn grid explosive area-of-effect damage, charred ash animation, and explosion particles.

Prompt 3.3.50

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-12T10:35:45+07:00
- **Prompt Content:**

/grill-me /goal

Next task: Update the range of explosion of the Explode-o-nut (the range where if a zombie's cer

- **Task Accomplished:** Implemented Explode-o-Nut entity with 3x3 lawn grid explosive area-of-effect damage, charred ash animation, and explosion particles.

Prompt 3.3.51

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-16T18:06:18+07:00
- **Prompt Content:**

/grill-me /goal

That's good. Now, next tasks, in Bowling Level:

- Add 2 more waves between wave 4 and wave 5. The first added wave is just the same as wave 4 w:

- **Task Accomplished:** Integrated wave spawner manager in `BowlingLevel.cpp` , scheduling zombie waves across 5 lanes with multi-hit armor damage states.

Prompt 3.3.52

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-18T21:47:57+07:00
- **Prompt Content:**

/grill-me

Now, next tasks, in Bowling level:

- Add a menu button on the top right corner of the window. The button should be SIMILAR to the 'Pause' button in the previous version.
 - Clicking it should spawn the menu dialog box, and pause the game. The menu dialog box is similar to the one in the previous version.
- Remember to generate test images for my approval.

Create an implementation plan, and wait for my approval.

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.53

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-18T21:54:10+07:00
- **Prompt Content:**

/grill-me

Tweak the pause menu dialog:

- Remove the "3D Acceleration" and "Full Screen" options.
- Move the labels, sliders and button textures in the menu dialog box downwards, so it fits perfectly.

Update the implementation plan, and wait for my approval.

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.54

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-18T21:56:58+07:00
- **Prompt Content:**

/grill-me

Generate new test images based on the implementation plan.

Update the implementation plan, and wait for my approval.

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.55

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-18T22:04:32+07:00
- **Prompt Content:**

/grill-me

Move the labels, sliders and button textures in the menu dialog box downwards, so it fits perfectly.

Generate new test images based on the implementation plan.

Update the implementation plan, and wait for my approval.

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.56

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-18T22:10:55+07:00
- **Prompt Content:**

/grill-me /goal

Fix the Menu button, the 3-slice button components don't align well here.

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.57

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-18T22:15:35+07:00
- **Prompt Content:**

/grill-me /goal

Fix this issue: The "BACK TO GAME" button in the pause menu dialog box doesn't show the label "I

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.58

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-18T22:20:46+07:00
- **Prompt Content:**

/graphify Summarize how the pause menu dialog was implemented in the bowling level.

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.59

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-18T22:24:23+07:00
- **Prompt Content:**

/grill-me

Now, next tasks, for the Level 1:

- You must read and analyze the code in git commit `b0449f66ee7e8879961823dd86839c4d10a95467`, ;
- Implement the exact same In-Game Menu & Pause Dialog menu for Level 1.

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.60

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-18T22:57:47+07:00
- **Prompt Content:**

/graphify Why right-clicking in bowling level and level 1 spawns the pole-vaulting zombie instead

- **Task Accomplished:** Integrated wave spawner manager in BowlingLevel.cpp , scheduling zombie waves across 5 lanes with multi-hit armor damage states.

Prompt 3.3.61

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-27T21:09:55+07:00
- **Prompt Content:**

/grill-me /graphify

Now, next task:

- Analyze how the label "Level 1" and the zombie progress bar in Level 1 is implemented.

- **Task Accomplished:** Integrated wave spawner manager in BowlingLevel.cpp , scheduling zombie waves across 5 lanes with multi-hit armor damage states.

Prompt 3.3.62

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-27T21:13:14+07:00
- **Prompt Content:**

/grill-me

Now, next tasks for Wall-nut Bowling level:

- Analyze how the label "Level 1" and the zombie progress bar in Level 1 is implemented.
 - Based on how the label "Level 1" and the zombie progress bar in Level 1 is implemented that you
- Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Integrated wave spawner manager in `BowlingLevel.cpp` , scheduling zombie waves across 5 lanes with multi-hit armor damage states.

Prompt 3.3.63

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-27T21:16:07+07:00
- **Prompt Content:**

/grill-me

Minor changes: Update the Level Label to be "Wall-nut Bowling" and use the "House of Terror" for Update the implementation plan first, and wait for my approval.

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.64

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T11:58:33+07:00
- **Prompt Content:**

/graphify List all zombies in the wall-nut bowling level.

- **Task Accomplished:** Integrated wave spawner manager in BowlingLevel.cpp , scheduling zombie waves across 5 lanes with multi-hit armor damage states.

Prompt 3.3.65

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T12:07:08+07:00
- **Prompt Content:**

/grill-me

Now, next tasks for Wall-nut Bowling level:

Make the Pole-vaulting zombie interact with the Normal Wall-nut as follows:

- First Contact (Vault Trigger): If the Pole-vaulting Zombie still has his pole, encountering a
- Nut Behavior: The normal Wall-nut does not bounce or deal damage; it passes directly underneath
- Subsequent Contacts: Vaulting strips the zombie of his pole and slows his movement to a standstill

The Pole-vaulting zombie interacts with the Explode-o-nut and Giant Wall-nut normally and MUST NOT

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Created Giant Wall-nut entity with heavy non-deflecting rolling mechanics that plow straight through zombies in its lane.

Prompt 3.3.66

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T12:22:18+07:00
- **Prompt Content:**

/grill-me /goal

Now, move the "Wall-nut Bowling" text label in the level upwards, because right now it goes over the level. Also, in the zombie wave progress bar, switch the zombie head icon and the flag icon.

- **Task Accomplished:** Integrated wave spawner manager in BowlingLevel.cpp , scheduling zombie waves across 5 lanes with multi-hit armor damage states.

Prompt 3.3.67

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T13:50:20+07:00
- **Prompt Content:**

/grill-me

Now, next tasks for bowling level:

I encounter a problem which is stated in @[temp/bowling_pole-vaulting_zombie_plan.md] . Read and

- **Task Accomplished:** Integrated wave spawner manager in BowlingLevel.cpp , scheduling zombie waves across 5 lanes with multi-hit armor damage states.

Prompt 3.3.68

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T16:43:53+07:00
- **Prompt Content:**

/graphify How is the simulation speed-up button in the levels from 1 to 6 implemented?

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.69

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T16:48:17+07:00
- **Prompt Content:**

/grill-me

For bowling level:

- Implement the simulation speed-up button EXACTLY like the levels from 1 to 6.
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.70

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T20:58:32+07:00
- **Prompt Content:**

/graphify What are the loss conditions in Level 1 to 6? What appears on the game screen when the

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.71

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T21:01:31+07:00
- **Prompt Content:**

/grill-me

Next tasks for Bowling Level:

- Change the coordinates of the house door in the loss condition of Bowling Level, to be EXACTLY
 - What Happens & Appears on Screen When the Player Loses: Make the staged loss sequence and the
- Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.72

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T21:10:56+07:00
- **Prompt Content:**

/graphify Is the gameover sequence in Bowling Level identical to Level 1 to 6 and Vasebreaker Le

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.73

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T21:12:57+07:00
- **Prompt Content:**

/graphify How are the lawnmowers implemented in Level 1 to 6?

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.74

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T21:15:32+07:00
- **Prompt Content:**

/grill-me

Next tasks for the Wall-nut Bowling level:

- Implement the lawnmowers for bowling level, EXACTLY like in the Levels 1 to 6.

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.75

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T21:24:47+07:00

- **Prompt Content:**

/graphify Are the lawnmowers in bowling level implemented identically to level 1 to 6?

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.76

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T21:28:33+07:00
- **Prompt Content:**

/graphify In levels 1 to 6, what happens in the game when the win condition is triggered?

- **Task Accomplished:** Implemented card pick-and-place cursor logic, restricting valid plant placement tiles strictly to columns 1–3 behind the red bowling line.

Prompt 3.3.77

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T21:34:07+07:00
- **Prompt Content:**

/grill-me

From the previous information:

- Implement The Victory Sequence (Audio, Progression, and Visuals) for Wall-nut Bowling level, I
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.78

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T11:31:45+07:00
- **Prompt Content:**

/graphify

What are the HP of each type of zombie in the Bowling level? What are the damage of each type of

- **Task Accomplished:** Integrated wave spawner manager in BowlingLevel.cpp , scheduling zombie waves across 5 lanes with multi-hit armor damage states.

Prompt 3.3.79

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T11:36:22+07:00
- **Prompt Content:**

/grill-me

Change the damage of the Bowling nuts in the bowling level:

- Normal Wall-nut: Hitting a Pole-Vaulting Zombie after vaulting, change from 1800 damage to the
- Giant Wall-nut: Change from 1000 damage to 1800 damage per contact.

Create a plan first

- **Task Accomplished:** Created Giant Wall-nut entity with heavy non-deflecting rolling mechanics that plow straight through zombies in its lane.

Prompt 3.3.80

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T11:38:37+07:00
- **Prompt Content:**

/grill-me

Update the implementation plan:

- Normal Wall-nut: Contact with Pole-Vaulting Zombie after vaulting now deals exactly 500HP.

- **Task Accomplished:** Integrated wave spawner manager in BowlingLevel.cpp , scheduling zombie waves across 5 lanes with multi-hit armor damage states.

Prompt 3.3.81

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T11:41:10+07:00
- **Prompt Content:**

/grill-me /goal

Minor changes: In the bowling level, the default of the Debug mode is ON, change it to OFF by default.

- **Task Accomplished:** Refined Wall-nut Bowling gameplay logic, coordinate centering, visual overlays, and interaction mechanics.

Prompt 3.3.82

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T22:25:39+07:00
- **Prompt Content:**

/grill-me

Task for the Bowling level:

- Play the looping BGM @[assets/sounds/wallnut_bowling_bgm.ogg] instead of the current day stage BGM.
Create an implementation plan, and wait for my approval.

- **Task Accomplished:** Integrated Wall-nut Bowling background music stream (assets/sounds/watery_grave.ogg) with volume controls.

3.4 Adventure Level Implementation

- **Conversation ID:** 947fe065-a531-400d-8838-1ef148039015
- **Category / Domain:** Adventure / Daytime Level 1
- **Primary Contributor:** Nguyễn Phúc Khánh

- **Date Range (UTC+7):** 2026-07-18 to 2026-09-01
- **Total Active Prompts:** 61

Prompt 3.4.1

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-18T16:26:16+07:00
- **Prompt Content:**

/teamwork-preview /grill-me /game-state-and-levels

Read and analyze the entire codebase.

Okay, now create a new screen where it's the daytime lawn level background in PvZ, with the plan
Create a plan first and wait for my approval.

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.2

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-18T16:38:46+07:00
- **Prompt Content:**

What is "Run ./build/pvz_reanim_viewer."? It should be "./build/PvZGame".

Also, be more patient and wait more time when compiling the project.

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.3

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-24T12:42:13+07:00
- **Prompt Content:**

/grill-me

Analyze the updated codebase, and the @[assets] folder. Your task now is to change the fonts in Create a plan first and wait for my approval.

- **Task Accomplished:** Engineered sun currency economy, implementing naturally falling sun items, sunflower production timers, and smooth bezier collection animations.

Prompt 3.4.4

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-24T12:53:22+07:00
- **Prompt Content:**

/grill-me

The Seed Packet Price Font is not `Pix118Bold`, but `Pico-129` instead. Also, use `assets/data/\_` Update the implementation plan first and wait for my approval.

- **Task Accomplished:** Integrated top seed bank UI bar with packet cooldown recharging sweeps and sun cost verification.

Prompt 3.4.5

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-24T12:55:23+07:00
- **Prompt Content:**

Read the rules and update the implementation plan.

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.6

- **Member:** Nguyễn Phúc Khánh

- **Sent At:** 2026-07-24T13:00:46+07:00
- **Prompt Content:**

/grill-me

There is a bug, the fonts are now black boxes and there are no real digits being displayed on the screen. Create the implementation plan first and wait for my approval.

- **Task Accomplished:** Engineered sun currency economy, implementing naturally falling sun items, sunflower production timers, and smooth bezier collection animations.

Prompt 3.4.7

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-24T13:09:48+07:00
- **Prompt Content:**

/grill-me /goal

The bug was fixed. Make the seed packet price text be centered horizontally, but vertically the

- **Task Accomplished:** Engineered sun currency economy, implementing naturally falling sun items, sunflower production timers, and smooth bezier collection animations.

Prompt 3.4.8

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-24T17:29:42+07:00
- **Prompt Content:**

/grill-me

From this reference image, reposition crazy dave and the license plate "Main Menu" button correctly. Create the implementation plan first and wait for my approval.

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.9

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-24T17:35:25+07:00
- **Prompt Content:**

/grill-me /goal

The position of crazy dave is still incorrect and is downward too much. I can only see his head. Fix his position.

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.10

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-24T17:38:52+07:00
- **Prompt Content:**

/grill-me /goal

The position of crazy dave is still incorrect and is horizontally centered too much. He should t Fix his position.

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.11

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-24T17:44:59+07:00
- **Prompt Content:**

/grill-me

Crazy dave's position looks okay now. Move the license plate "Main Menu" button a little bit to

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.12

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-24T17:54:00+07:00
- **Prompt Content:**

/grill-me /goal

Change the "MAIN MENU" text label's color to `rgba(102, 152, 235)`. Also increase the scale of 1

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.13

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-27T10:35:47+07:00
- **Prompt Content:**

/grill-me

Next, read and analyze the updated codebase. Then, for the Level 1 screen:

- Create a plant seed preselection screen
 - I can click on unselected seed packets and they get selected, I must also be able to unselect
 - There is a confirm button, if i click it, it takes me to the usual level 1 gameplay screen
- Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Integrated top seed bank UI bar with packet cooldown recharging sweeps and sun cost verification.

Prompt 3.4.14

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-27T10:51:39+07:00

- **Prompt Content:**

/grill-me

Clarification: Clicking "Start Adventure" starts up plant reanimation testing screen, that's for
Update the implementation plan first, and wait for my approval.

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.15

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-27T11:00:19+07:00
- **Prompt Content:**

/grill-me From the implementation plan, review this file @[temp/preselect-design.md] .

Combine that file to create a better design for the "Plant Seed Preselection Screen for Level 1"
Update the implementation plan first, and wait for my approval.

- **Task Accomplished:** Integrated top seed bank UI bar with packet cooldown recharging sweeps and sun cost verification.

Prompt 3.4.16

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-27T17:00:39+07:00
- **Prompt Content:**

/grill-me

Next task:

There is a "seed packet" in the preselection screen that doesn't look like a seed packet image
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Integrated top seed bank UI bar with packet cooldown recharging sweeps and sun cost verification.

Prompt 3.4.17

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-27T22:10:55+07:00
- **Prompt Content:**

/grill-me

Next task:

In the seed packet preselection menu, there were only 4 columns of packets, leaving the right half empty. Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Integrated top seed bank UI bar with packet cooldown recharging sweeps and sun cost verification.

Prompt 3.4.18

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-27T22:22:26+07:00
- **Prompt Content:**

/grill-me /goal

Next task:

In "Level 1", the sun counter text label's position during the seed preselection menu and during the sunflower production menu is inconsistent. Find the root cause. Fix the UI bug.

- **Task Accomplished:** Engineered sun currency economy, implementing naturally falling sun items, sunflower production timers, and smooth bezier collection animations.

Prompt 3.4.19

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-27T22:36:08+07:00
- **Prompt Content:**

/grill-me

Next task:

In Level 1, during seed selection phase, the text labels of the sun costs for the seed packets :
I sent you an image of the UI bug.
Fix the root cause for this UI bug. Fix the UI bug.
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Engineered sun currency economy, implementing naturally falling sun items, sunflower production timers, and smooth bezier collection animations.

Prompt 3.4.20

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-27T22:45:37+07:00
- **Prompt Content:**

/grill-me /goal

Tiny change: Don't use the `getPlantCost` lambda function with multiple if statements. Create a

- **Task Accomplished:** Engineered sun currency economy, implementing naturally falling sun items, sunflower production timers, and smooth bezier collection animations.

Prompt 3.4.21

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-29T14:25:31+07:00
- **Prompt Content:**

/grill-me /goal

Next task:

- For the "Let's Rock" button: When the button is NOT disabled: If the mouse hovers on it, disp.

- **Task Accomplished:** Integrated top seed bank UI bar with packet cooldown recharging sweeps and sun cost verification.

Prompt 3.4.22

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-29T14:36:14+07:00
- **Prompt Content:**

/grill-me /goal

Tiny formatting task:

- The "Let's Rock" text's font changes to the "Dwarven Todcraft" font. The text's size must be 6
- The text must also be grey when the button is disabled, white when button is NOT disabled and

- **Task Accomplished:** Integrated top seed bank UI bar with packet cooldown recharging sweeps and sun cost verification.

Prompt 3.4.23

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-29T14:48:33+07:00
- **Prompt Content:**

/grill-me

Next task:

- There is a UI bug. I sent you a reference image. The chosen seed packets in the selection menu
- Find the root cause of the bug and fix it.
- Create an implementation plan first, and wait for approval.

- **Task Accomplished:** Integrated top seed bank UI bar with packet cooldown recharging sweeps and sun cost verification.

Prompt 3.4.24

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-29T15:08:50+07:00
- **Prompt Content:**

/grill-me /goal

Tiny formatting task:

Move the selected seed packets in the selection menu to the left so that the spacings between 16

- **Task Accomplished:** Integrated top seed bank UI bar with packet cooldown recharging sweeps and sun cost verification.

Prompt 3.4.25

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-29T15:24:54+07:00
- **Prompt Content:**

/grill-me /goal

Now, next task:

- In the seed bank of the seed selection menu, display the 7 seed packet silhouettes @[assets/ir

- **Task Accomplished:** Integrated top seed bank UI bar with packet cooldown recharging sweeps and sun cost verification.

Prompt 3.4.26

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-29T15:35:44+07:00
- **Prompt Content:**

/grill-me

Now, next task:

- In the seed chooser background @[assets/images/SeedChooser_Background.png] of the preselection
- The title text should also be moved and scaled to be at the center of the top of the @[assets,
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Integrated top seed bank UI bar with packet cooldown recharging sweeps and sun cost verification.

Prompt 3.4.27

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-29T15:41:34+07:00
- **Prompt Content:**

/grill-me /goal

Tiny formatting change:

Move the title text, like in the image, to be vertically centered in the seed chooser panel top

- **Task Accomplished:** Implemented initial level camera panning sequence showing the right side of the lawn before sliding back to the player's lawn.

Prompt 3.4.28

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-11T20:55:54+07:00
- **Prompt Content:**

/graphify What is damage of cherry bomb and jalapeno

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.29

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-12T21:41:16+07:00
- **Prompt Content:**

/grill-me

Now, next task:

Task: Level 1 Seed Preselection & Camera/Background Transition Feature

We need to implement a dynamic camera pan and background transition sequence for the start of L

Requirements Specification:

1. **Initial Seed Preselection View:**

- When launching Level 1, position the camera/viewport on the **rightmost side of the lawn st**
- Keep the seed packet preselection UI active and interactive over this background view.

2. **Transition Sequence:**

- Triggered immediately after the player finishes pre-selecting their seeds and clicks "Let's
- Smoothly pan/transition the background viewport from the street view horizontally across to

3. **Technical & Architectural Considerations:**

- Research best practices for smooth 2D camera scrolling/lerping and background state transi
- Handle asset positioning boundaries, UI layer locking/fading during movement, and event sta

Expected Output:

- Create and present a detailed **Implementation Plan** artifact first.
- **Do not modify or write code yet.** Wait for my explicit review and approval of the plan befo

- **Task Accomplished:** Implemented initial level camera panning sequence showing the right side of the lawn before sliding back to the player's lawn.

Prompt 3.4.30

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-12T22:18:20+07:00
- **Prompt Content:**

/grill-me

Now, next task:

Task: Implement the **Zombie Preview** feature during the seed preselection phase for Level 1:

Technical Requirements

1. **Wave Data Parsing**: Inspect Level 1's wave configuration data to extract all unique zombie types.
2. **Preview Zombie Generation**: Instantiate sample preview zombie entities for each unique type.
3. **Animation & Behavior**: Render these preview zombies playing their `IDLE` animation loop.
4. **State Machine Hook**: Keep preview rendering contained within `SeedSelection` and `PanToLawn` states.

Create an implementation plan first, and wait for my approval.

- **Task Accomplished**: Implemented initial level camera panning sequence showing the right side of the lawn before sliding back to the player's lawn.

Prompt 3.4.31

- **Member**: Nguyễn Phúc Khánh
- **Sent At**: 2026-08-12T22:33:36+07:00
- **Prompt Content**:

/grill-me

Fix these bugs:

- Bug 1: The zombie preview entities are too far to the right compared to the streets.
 - Bug 2: During `PanToLawn` phase, the zombie previews stay still compared to the game window, they don't follow the camera.
- Find the root cause of these bugs, and create an implementation plan to fix them.

- **Task Accomplished**: Implemented initial level camera panning sequence showing the right side of the lawn before sliding back to the player's lawn.

Prompt 3.4.32

- **Member**: Nguyễn Phúc Khánh
- **Sent At**: 2026-08-12T22:59:01+07:00
- **Prompt Content**:

/grill-me /goal

Bug: The zombie preview entities are too far to the left compared to the streets.

Tiny fix: The street region is actually located between `x = 1130` and `1350`.

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.33

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-15T22:13:17+07:00
- **Prompt Content:**

/graphify What is the zombie wave layout in each of Level 1 and Bowling level?

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.34

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-15T22:24:17+07:00
- **Prompt Content:**

/grill-me

Now, next tasks, in Level 1:

- Add 2 more waves between wave 4 and wave 5. The first added wave is just the same as wave 4 with 10 zombies.
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.35

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-18T23:02:39+07:00
- **Prompt Content:**

/graphify How was the pause "Menu" button and the menu dialog box implemented in Level 1 game scene?

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.36

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-18T23:06:05+07:00
- **Prompt Content:**

/grill-me

Based on how the ****In-Game Pause Dialog Box**** was implemented in Level 1, do these tasks:

- Add the pause Menu button and the In-Game Pause Dialog Box even during the seed preselection routine.
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Implemented initial level camera panning sequence showing the right side of the lawn before sliding back to the player's lawn.

Prompt 3.4.37

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T16:34:39+07:00
- **Prompt Content:**

/grill-me /goal

Next tasks for Level 1 to 6:

- Change the font of the level label "Level x" to use the "House of Terror" font, like the Vase of
- Move the label so it stays in the game window, and doesn't overlap with the zombie wave progress

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.38

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T21:53:01+07:00
- **Prompt Content:**

/grill-me /goal

Change the simulation speed button in all levels (Level 1 to 6, bowling level, vasebreaker level)

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.39

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T11:35:59+07:00
- **Prompt Content:**

/graphify What levels (among the levels 1 to 6, wall-nut bowling, vasebreaker levels) have the 'Ready, Set, PLANT!' countdown overlay banner at the start of Level 1.

- **Task Accomplished:** Implemented animated 'Ready, Set, PLANT!' countdown overlay banner at the start of Level 1.

Prompt 3.4.40

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T12:34:38+07:00
- **Prompt Content:**

/grill-me

There is a bug in the "Ready, Set" countdown in all levels (level 1 to 6, bowling, vasebreaker):

- The countdown goes "Ready", "Set", then "Plant" or "Break", as expected. But then it loops back to "Ready".

Find the root cause of this bug, and create an implementation plan to fix it.

- **Task Accomplished:** Implemented animated 'Ready, Set, PLANT!' countdown overlay banner at the start of Level 1.

Prompt 3.4.41

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T16:29:46+07:00
- **Prompt Content:**

/graphify Explain all gameplay details about the Levels 1 to 6.

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.42

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T16:42:49+07:00
- **Prompt Content:**

/grill-me

Next tasks, for Level 1 to 6:

- Fix what zombies should be displayed for preview in the seed selection menu, based on the type of seed.

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Integrated top seed bank UI bar with packet cooldown recharging sweeps and sun cost verification.

Prompt 3.4.43

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T16:59:07+07:00
- **Prompt Content:**

/grill-me

In the levels 1 to 6:

- The GraveBuster plant can be placed on lawn tiles that are not occupied by graves.

Find the root cause of the bug, and create an implementation plan to fix it.

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.44

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T21:38:19+07:00
- **Prompt Content:**

/grill-me

Fix this bug: In levels 1 to 6, when a zombie meets a garlic plant, the zombie doesn't show the I want the zombie to eat the garlic just ONCE, with the eating animation, dealing a bit of damage. Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.45

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T21:59:48+07:00
- **Prompt Content:**

/grill-me

There is a bug: On the lawn, when I place a plant on leftmost column, and the Pole-vaulting zombie comes, the zombies do not move.
Find the root cause of the bug, and create an implementation plan to fix it.

- **Task Accomplished:** Implemented lawnmower defense triggers at the left lawn boundary, clearing all zombies in the triggered lane upon breach.

Prompt 3.4.46

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T11:00:05+07:00
- **Prompt Content:**

/grill-me

Minor changes:

In the victory screen and loss screen of the Levels 1 to 6, the Wall-nut Bowling level, and the Vasebreaker level:
- Change the font of the big "Level Completed" to the `HouseOfTerror` font.
- Change the font of all the smaller text to the `BrianneTod` font.
- Also, in the victory screen, add a smaller text specifying how much cash the player gains after the level.
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.47

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T11:15:40+07:00
- **Prompt Content:**

/grill-me

Minor changes:

In the loss screen of the Levels 1 to 6, the Wall-nut Bowling level, and the Vasebreaker level:
- Move the "THE ZOMBIE ATE YOUR BRAINS" image to be centered on the window.
- Move the "Click anywhere to try again" a little bit lower so it does not overlap with the "THE ZOMBIE ATE YOUR BRAINS" image.
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.48

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T11:26:50+07:00
- **Prompt Content:**

/graphify When the last zombie dies in a level, does the victory screen play immediately, or does it wait for a short time before playing?

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.49

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T12:12:04+07:00
- **Prompt Content:**

/grill-me

Minor changes for Level 1 to 6:

- Set default sun counter when starting a level to 0 sun.

- **Task Accomplished:** Engineered sun currency economy, implementing naturally falling sun items, sunflower production timers, and smooth bezier collection animations.

Prompt 3.4.50

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T13:38:28+07:00
- **Prompt Content:**

/grill-me

Change the cash rewards of the Level 1 to 6, Wallnut Bowling level, and Vasebreaker level to match the Shop economy.
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.51

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T13:44:57+07:00
- **Prompt Content:**

/grill-me /goal

Also update the cash reward of the Brain Buster level too, to match the Shop economy.

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.52

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T14:07:00+07:00
- **Prompt Content:**

/grill-me

Does the day level drop sun from the sky naturally? If not, create an implementation plan to implement it.

- **Task Accomplished:** Engineered sun currency economy, implementing naturally falling sun items, sunflower production timers, and smooth bezier collection animations.

Prompt 3.4.53

- **Member:** Nguyễn Phúc Khánh

- **Sent At:** 2026-09-01T16:37:29+07:00
- **Prompt Content:**

/grill-me

There is a bug: When I play a daytime level, I don't see any sun entity dropping. Also, the 10-
Find the root cause of these bugs, and create an implementation plan to fix them.

- **Task Accomplished:** Engineered sun currency economy, implementing naturally falling sun items, sunflower production timers, and smooth bezier collection animations.

Prompt 3.4.54

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T17:26:57+07:00
- **Prompt Content:**

/grill-me

Add this feature: In the levels 1 to 6, and in the Wall-nut Bowling level:

- Right before the final wave starts (with the intense final wave SFX), show the @[assets/reanir
 - Right before an intense wave with a Flag Zombie (that is NOT a final wave), show the text `A t
- Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.55

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T17:59:57+07:00
- **Prompt Content:**

/grill-me /goal

Minor changes: Set the default starting sun counter of each and every level to 50, instead of 0

- **Task Accomplished:** Engineered sun currency economy, implementing naturally falling sun items, sunflower production timers, and smooth bezier collection animations.

Prompt 3.4.56

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T19:53:15+07:00
- **Prompt Content:**

/graphify

Are the victory and game over screens in Level 1 to 6 identical to those of the Bowling level ar

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.57

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T19:55:42+07:00
- **Prompt Content:**

/grill-me /goal

Among levels 4, 5, and 6, change the victory screen subtitle text of the level with no fog to ""

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.58

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T19:57:49+07:00
- **Prompt Content:**

/grill-me /goal

Among levels 4, 5, and 6, change the victory screen subtitle text of the level with fog to "You

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.59

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T21:14:28+07:00
- **Prompt Content:**

/graphify Search in @[assets] the music for the plant preselection menu in the adventure levels.

- **Task Accomplished:** Developed daytime lawn level (Level 1) gameplay features, grid collision hitboxes, and state management.

Prompt 3.4.60

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T21:54:01+07:00
- **Prompt Content:**

/grill-me

Task for the adventure levels (level 1 to 6):

- Play the looping music @[assets/sounds/choose_your_seeds.ogg] during the plant preselection menu.
Create an implementation plan, and wait for my approval.

- **Task Accomplished:** Integrated top seed bank UI bar with packet cooldown recharging sweeps and sun cost verification.

Prompt 3.4.61

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T21:56:27+07:00
- **Prompt Content:**

/grill-me

Update the implemetation plan:

- Even during the camera panning to the lawn, the @[assets/sounds/choose_your_seeds.ogg] BGM st:
- **Task Accomplished:** Implemented initial level camera panning sequence showing the right side of the lawn before sliding back to the player's lawn.

3.5 Compilation Bug Fixer

- **Conversation ID:** 5f68ae79-789e-4038-a502-7e4a28d61d45
- **Category / Domain:** Build Fixes & Conflict Resolution
- **Primary Contributor:** Nguyễn Phúc Khánh & Từ Hoàng Anh
- **Date Range (UTC+7):** 2026-07-27 to 2026-09-01
- **Total Active Prompts:** 11

Prompt 3.5.1

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-27T23:04:16+07:00
- **Prompt Content:**

/grill-me

Follow @[.agents/rules/AGENTS.md] . Read and analyze the entire codebase.

Your task:

The code returns an error when compiling. Fix that error

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Fixed compilation errors, linker unresolved symbols, and build configuration warnings.

Prompt 3.5.2

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-18T23:14:56+07:00
- **Prompt Content:**

/grill-me

Read and analyze the codebase carefully. Then diagnose this issue:

- When I run the PvZGame application and mess around with it for a while, the app suddenly crash
Find the root cause of this issue.

- **Task Accomplished:** Fixed compilation errors, linker unresolved symbols, and build configuration warnings.

Prompt 3.5.3

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-19T14:09:15+07:00
- **Prompt Content:**

/grill-me

What's the best fix that changes the least of the structure of codebase?

- **Task Accomplished:** Fixed compilation errors, linker unresolved symbols, and build configuration warnings.

Prompt 3.5.4

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-19T14:25:29+07:00
- **Prompt Content:**

/grill-me

Are all the problems in @[temp/log.txt] fixed now?

- **Task Accomplished:** Fixed compilation errors, linker unresolved symbols, and build configuration warnings.

Prompt 3.5.5

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-22T12:14:43+07:00
- **Prompt Content:**

/graphify /grill-me

Read and analyze the codebase carefully. Then diagnose this issue:

- When I run the PvZGame application, and go to Vasebreaker level, and mess around with the menu
Find the root cause of this issue, and create an implementation plan to fix it.

- **Task Accomplished:** Fixed compilation errors, linker unresolved symbols, and build configuration warnings.

Prompt 3.5.6

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-22T12:28:57+07:00
- **Prompt Content:**

/graphify /grill-me

Read and analyze the codebase carefully. Then diagnose this issue:

- When I run the PvZGame application, and go to Vasebreaker level, and mess around with the menu
Find the root cause of this issue, and UPDATE the implementation plan to fix it.

- **Task Accomplished:** Fixed compilation errors, linker unresolved symbols, and build configuration warnings.

Prompt 3.5.7

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-22T12:40:49+07:00
- **Prompt Content:**

/grill-me

Are all the problems in @[temp/log2.txt] fixed now?

- **Task Accomplished:** Fixed compilation errors, linker unresolved symbols, and build configuration warnings.

Prompt 3.5.8

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-22T17:52:37+07:00
- **Prompt Content:**

/grill-me /goal

Analyze the errors in @[temp/log3.txt] during GitHub Actions CI/CD, and provide a quick fix for

- **Task Accomplished:** Fixed compilation errors, linker unresolved symbols, and build configuration warnings.

Prompt 3.5.9

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-22T18:00:00+07:00
- **Prompt Content:**

/grill-me /goal

Solve this quick error for GitHub Actions:

/home/runner/work/CS202-GameProject/CS202-GameProject/src/core/resources.cpp: In member function 'Texture2D Resources::GetTexture(const std::string&) const':

/home/runner/work/CS202-GameProject/CS202-GameProject/src/core/resources.cpp:222:9: error: 'strncmp' was not declared in this scope

```
222 | if (strncmp(start, "IMAGE_REANIM_", 13) == 0) {
    | ^~~~~~
```

/home/runner/work/CS202-GameProject/CS202-GameProject/src/core/resources.cpp:7:1: note: 'strncmp' is defined in header ' '; did you forget to ' #include ' ?

```
6 | #include
```

```
+++ | + #include
```

```
7 |
```

- **Task Accomplished:** Resolved Raylib image format loading warnings and updated asset loading paths.

Prompt 3.5.10

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T23:05:06+07:00
- **Prompt Content:**

/grill-me

After merging in git, there is a merge conflict.

Inspect the codebase carefully and carefully resolve the merge conflict.

- **Task Accomplished:** Resolved Git merge conflicts across header files and restored clean compilation across all build targets.

Prompt 3.5.11

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T21:24:47+07:00
- **Prompt Content:**

/grill-me

New task:

- After merging in git, there is a merge conflict.
- Inspect the git history carefully, and carefully resolve the merge conflict.

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Resolved Git merge conflicts across header files and restored clean compilation across all build targets.

3.6 Making Shop In Game

- **Conversation ID:** de4b6fd4-43fc-4aef-9ddd-6c6764b28327
- **Category / Domain:** Crazy Dave's Shop UI & Currency
- **Primary Contributor:** Nguyễn Phúc Khánh
- **Date Range (UTC+7):** 2026-07-23 to 2026-09-01
- **Total Active Prompts:** 35

Prompt 3.6.1

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-23T20:10:49+07:00
- **Prompt Content:**

/grill-me Analyze the entire codebase. Your task is to change the default resolution to the "Cor
Create an implementation plan first, and wait for my approval. You can ask me questions

- **Task Accomplished:** Developed Crazy Dave's Shop UI elements, coordinate alignment, button hover states, and audio triggers.

Prompt 3.6.2

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-23T22:06:44+07:00
- **Prompt Content:**

/grill-me Analyze the entire codebase and @[assets] folder. Add a Shop button in the main menu
Create an implementation plan first, and wait for my approval. You can ask me questions.

- **Task Accomplished:** Developed Crazy Dave's Shop UI elements, coordinate alignment, button hover states, and audio triggers.

Prompt 3.6.3

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-23T22:12:16+07:00

- **Prompt Content:**

/grill-me

Question 1: The Shop button is located exactly where it is like in the original PvZ game on the

Question 2: The "Main Menu" button is located exactly where it is like in the original PvZ game

Update implementation plan first, and wait for my approval. You can ask me questions.

- **Task Accomplished:** Designed and rendered Crazy Dave's Shop (ShopMenu), loading Crazy Dave sprite, car trunk background, and shelf item slots.

Prompt 3.6.4

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-24T11:17:39+07:00
- **Prompt Content:**

/grill-me

In the Shop menu, add crazy dave and his car, and the seed packets. Clicking the seed packets do

Create an implementation plan first, and wait for my approval. You can ask me questions.

- **Task Accomplished:** Designed and rendered Crazy Dave's Shop (ShopMenu), loading Crazy Dave sprite, car trunk background, and shelf item slots.

Prompt 3.6.5

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-24T11:36:25+07:00
- **Prompt Content:**

/grill-me

Can you search up on the Internet to find the `(x, y)` coordinates on the 800x600 canvas of the

Question 3: Choose random seed packet assets that my teammate added.

Use alpha-masking to get transparent background similar to what's in the codebase, and use pixel

Update implementation plan first, and wait for my approval. You can ask me questions while work:

- **Task Accomplished:** Developed Crazy Dave's Shop UI elements, coordinate alignment, button hover states, and audio triggers.

Prompt 3.6.6

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-24T12:26:39+07:00
- **Prompt Content:**

/grill-me /goal

Apply alpha-masking for @[assets/images/Store_Car.jpg] and the mask@[assets/images/Store_Car_1.t

- **Task Accomplished:** Designed and rendered Crazy Dave's Shop (ShopMenu), loading Crazy Dave sprite, car trunk background, and shelf item slots.

Prompt 3.6.7

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-24T21:16:05+07:00
- **Prompt Content:**

/grill-me

Analyze the updated codebase.

In the Shop menu, for the 8 seed packets, reposition them like so:

4 cards on first row:

- First card: `(x,y)` coordinates are: top left `(186\*2, 155\*2)`, bottom right `(211\*2, 190\*2)`.
- Second card: top left `(223\*2, 155\*2)`, bottom right `(248\*2, 190\*2)`.
- 3rd and 4th cards follow the same row and same spacing.

4 cards on second row:

- First card: `(x,y)` coordinates are: top left `(210\*2, 103\*2)`, bottom right `(235\*2, 138\*2)`.
- Second card: top left `(247\*2, 103\*2)`, bottom right `(272\*2, 139\*2)`.
- 3rd and 4th cards follow the same row and same spacing.

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Designed and rendered Crazy Dave's Shop (ShopMenu), loading Crazy Dave sprite, car trunk background, and shelf item slots.

Prompt 3.6.8

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-24T21:31:48+07:00
- **Prompt Content:**

/grill-me

Next, in the Shop menu, there may be multiple pages, each page has 8 items like the 8 seed packs

- If it is the very first page, the "prev" button is disabled. Else, "prev" button is active, and has
- If it is the final page, the "next" button is disabled. Else, "next" button is active, and has

Use this image as a reference image for the 2 buttons. For the assets, use @[assets/images/Store

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Implemented multi-page catalog pagination with state-driven PREV / NEXT navigation buttons and hover highlights.

Prompt 3.6.9

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-24T21:40:44+07:00
- **Prompt Content:**

/grill-me

There is a UI bug: The "PREV" and "NEXT" buttons' positions are incorrect, they are located further

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Designed and rendered Crazy Dave's Shop (ShopMenu), loading Crazy Dave sprite, car trunk background, and shelf item slots.

Prompt 3.6.10

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-24T21:56:16+07:00
- **Prompt Content:**

/grill-me /goal

Revert ONLY the pageTextBounds coordinates back.

There is a UI bug: The "PREV" and "NEXT" buttons' positions are incorrect.

Do this fix: Translate the prev button subtract `x` coordinate by 3, subtract `y` coordinate 14

Do the math correctly.

- **Task Accomplished:** Implemented multi-page catalog pagination with state-driven PREV / NEXT navigation buttons and hover highlights.

Prompt 3.6.11

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-24T21:59:10+07:00
- **Prompt Content:**

No, I mean subtract from the last change you did, for example prevRect was `254.0f, 474.0f`, and

- **Task Accomplished:** Implemented multi-page catalog pagination with state-driven PREV / NEXT navigation buttons and hover highlights.

Prompt 3.6.12

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-24T22:18:46+07:00
- **Prompt Content:**

/grill-me /goal

Okay, next, in the Shop menu, move Crazy Dave's position by -35 pixels in x-coordinate.

- **Task Accomplished:** Designed and rendered Crazy Dave's Shop (ShopMenu), loading Crazy Dave sprite, car trunk background, and shelf item slots.

Prompt 3.6.13

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-28T13:09:46+07:00
- **Prompt Content:**

/grill-me

Next task:

- Whenever I click on a seed packet in the Shop menu, it should immediately be sold. The seed packet should be sold immediately. Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Rendered PopCap price tags (Store_PriceTag.png) using BrianneTod and Pico-129 custom bitmap font glyphs.

Prompt 3.6.14

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-28T17:39:56+07:00
- **Prompt Content:**

/grill-me

Next task:

- For each item in the Shop menu, display a price tag. The price tag is just like the original item. Create an implementation plan and wait for my approval.

- **Task Accomplished:** Rendered PopCap price tags (Store_PriceTag.png) using BrianneTod and Pico-129 custom bitmap font glyphs.

Prompt 3.6.15

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-28T17:46:13+07:00
- **Prompt Content:**

/grill-me

There is a UI bug: I see the price tag, but I don't see the price font text label of each seed ;
Find the root cause of the bug and fix it.

- **Task Accomplished:** Rendered PopCap price tags (Store_PriceTag.png) using BrianneTod and Pico-129 custom bitmap font glyphs.

Prompt 3.6.16

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-28T17:52:47+07:00
- **Prompt Content:**

/grill-me /goal

Re-read and follow @[.agents/rules/AGENTS.md] .

Do this minor fix: Center the price text horizontally AND vertically inside the price tag `STORE`

- **Task Accomplished:** Rendered PopCap price tags (Store_PriceTag.png) using BrianneTod and Pico-129 custom bitmap font glyphs.

Prompt 3.6.17

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-28T19:55:45+07:00
- **Prompt Content:**

/grill-me

Now, next tasks:

- Read and analyze the codebase again, and analyze and follow @[.agents/rules/AGENTS.md] again.
 - Add a money counter in the Shop menu, with a money variable.
 - Display the money counter using @[assets/images/coinbank.png] , in the bottom right corner of
 - Initially, the player has `\$1,000,000`.
 - When I buy a seed packet, the money counter will be updated and decrease based on the price of
- Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Rendered PopCap price tags (Store_PriceTag.png) using BrianneTod and Pico-129 custom bitmap font glyphs.

Prompt 3.6.18

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-28T20:01:16+07:00
- **Prompt Content:**

/grill-me /goal

Tiny formatting tasks:

The text label for money counter should be colored green, using right alignment with small padding.

- **Task Accomplished:** Implemented player currency bank (coinbank.png starting at \$1,000,000) with real-time deductions upon purchasing seed packets.

Prompt 3.6.19

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-28T20:03:31+07:00
- **Prompt Content:**

/grill-me /goal

Tiny formatting tasks:

For the money counter text label, you still forgot to include the commas, like `\$1,000,000`. Fix it.

- **Task Accomplished:** Implemented player currency bank (coinbank.png starting at \$1,000,000) with real-time deductions upon purchasing seed packets.

Prompt 3.6.20

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-28T22:19:29+07:00
- **Prompt Content:**

/grill-me /goal

Tiny formatting task:

Change the color of the money counter's text label to `rgba(181, 246, 90)`.

- **Task Accomplished:** Implemented player currency bank (coinbank.png starting at \$1,000,000) with real-time deductions upon purchasing seed packets.

Prompt 3.6.21

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T10:26:22+07:00
- **Prompt Content:**

/grill-me

Fix the zombie icon buttons in the Zombie Almanac Menu:

- The border of the zombie icon buttons must have @[assets/images/Almanac_ZombieWindow2.png] .
 - For each zombie, get their still image (no animation at all), and place the zombie's upper half.
- Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Developed Crazy Dave's Shop UI elements, coordinate alignment, button hover states, and audio triggers.

Prompt 3.6.22

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T10:32:50+07:00
- **Prompt Content:**

/grill-me Update the implementation plan:

- Empty slots render `Almanac\_ZombieBlank.png` but don't render `Almanac\_ZombieWindow2.png`.

- **Task Accomplished:** Developed Crazy Dave's Shop UI elements, coordinate alignment, button hover states, and audio triggers.

Prompt 3.6.23

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T10:41:48+07:00
- **Prompt Content:**

/grill-me /goal

Fix this mistake: The zombie icon border image `Almanac\_ZombieWindow2.png` has some alpha transparency. Find the root cause and fix this mistake.

- **Task Accomplished:** Developed Crazy Dave's Shop UI elements, coordinate alignment, button hover states, and audio triggers.

Prompt 3.6.24

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T10:49:37+07:00
- **Prompt Content:**

/grill-me /goal

Fix these bugs in zombie almanac:

- The zombie icon border image `Almanac\_ZombieWindow2.png` has some alpha transparency, but the
 - The zombies' positions in the border are further to the left. I want you to move them and align them.
- Find the root cause and fix these bugs.

- **Task Accomplished:** Developed Crazy Dave's Shop UI elements, coordinate alignment, button hover states, and audio triggers.

Prompt 3.6.25

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T10:56:00+07:00
- **Prompt Content:**

/grill-me /goal

For the zombie icon buttons: Also draw the @[assets/images/Almanac_ZombieWindow.png] .

And also, the still zombie images protrudes the zombie icon border. Cut the zombie images to size.

- **Task Accomplished:** Developed Crazy Dave's Shop UI elements, coordinate alignment, button hover states, and audio triggers.

Prompt 3.6.26

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T11:01:48+07:00
- **Prompt Content:**

/grill-me /goal

Now, next tasks for zombie almanac icon buttons:

- Move the zombie almanac icon buttons downwards (only change their y-coordinates) to be vertically aligned.

- **Task Accomplished:** Designed and rendered Crazy Dave's Shop (ShopMenu), loading Crazy Dave sprite, car trunk background, and shelf item slots.

Prompt 3.6.27

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T11:10:03+07:00
- **Prompt Content:**

/grill-me

Next tasks for the Zombie Almanac menu:

- Place the day/night lawn background image to the background of the zombie info card, like the one in the reference image.
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Designed and rendered Crazy Dave's Shop (ShopMenu), loading Crazy Dave sprite, car trunk background, and shelf item slots.

Prompt 3.6.28

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T11:19:06+07:00
- **Prompt Content:**

/grill-me

Next tasks for the Zombie Almanac menu:

- Edit the FlagZombie entity to be exactly like the flag zombie in the testing menu and in the :
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Implemented multi-page catalog pagination with state-driven PREV / NEXT navigation buttons and hover highlights.

Prompt 3.6.29

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T11:25:49+07:00
- **Prompt Content:**

/graphify In the Zombie Almanac menu, are all the zombie preview entities exactly like the zomb:

- **Task Accomplished:** Implemented multi-page catalog pagination with state-driven PREV / NEXT navigation buttons and hover highlights.

Prompt 3.6.30

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T22:01:13+07:00
- **Prompt Content:**

/graphify Name all plants in the Shop menu which are implemented and which are not implemented.

- **Task Accomplished:** Developed Crazy Dave's Shop UI elements, coordinate alignment, button hover states, and audio triggers.

Prompt 3.6.31

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T22:59:48+07:00
- **Prompt Content:**

/graphify Name all plants in the Shop menu which are implemented and which are not implemented.

- **Task Accomplished:** Developed Crazy Dave's Shop UI elements, coordinate alignment, button hover states, and audio triggers.

Prompt 3.6.32

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T23:18:39+07:00
- **Prompt Content:**

/grill-me

In the Shop menu:

- Delete the seed packets of the plants that are NOT implemented in the game.
 - Add the seed packets of Snow Pea, Repeater, Chomper, Caltrop/Spikeweed, Spikerock, Gravebuster
- Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Developed Crazy Dave's Shop UI elements, coordinate alignment, button hover states, and audio triggers.

Prompt 3.6.33

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T13:34:46+07:00
- **Prompt Content:**

/grill-me

Change the seed packet costs in the Shop menu to match with the usefulness and strengths of each
Create an implementation plan, and wait for my approval.

- **Task Accomplished:** Developed Crazy Dave's Shop UI elements, coordinate alignment, button hover states, and audio triggers.

Prompt 3.6.34

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T17:13:42+07:00
- **Prompt Content:**

/grill-me

Does the Shop menu have this feature yet: Negative buzzer sound effect when purchasing without a grill.
If not, create an implementation plan to add this feature.

- **Task Accomplished:** Developed Crazy Dave's Shop UI elements, coordinate alignment, button hover states, and audio triggers.

Prompt 3.6.35

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T18:01:38+07:00
- **Prompt Content:**

/grill-me

Add this sound effect in Shop menu: When clicking PREV and NEXT buttons, play the `buttonclick.wav`.
Create an implementation plan first.

- **Task Accomplished:** Implemented multi-page catalog pagination with state-driven PREV / NEXT navigation buttons and hover highlights.

3.7 Options Menu Implementation Plan

- **Conversation ID:** 1187c4e1-e783-427b-ba46-6d45f0d82cde
- **Category / Domain:** Options Menu & Audio Controls
- **Primary Contributor:** Nguyễn Phúc Khánh

- **Date Range (UTC+7):** 2026-07-25 to 2026-09-01
- **Total Active Prompts:** 8

Prompt 3.7.1

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-25T17:37:19+07:00
- **Prompt Content:**

/grill-me

I want that when dragging the volume slider, it will change the volume immediately without click.
Also, read @[.agents/rules] to understand more.
Update the implementation plan first, and wait for my approval.

- **Task Accomplished:** Implemented real-time audio volume sliders in Options menu connected to AudioManager with drag sound effects.

Prompt 3.7.2

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-25T17:57:20+07:00
- **Prompt Content:**

/grill-me /goal

UI bug: Move the "OPTIONS" title and the music volume slider downwards. They are too high. The i
Fix this bug.

- **Task Accomplished:** Implemented real-time audio volume sliders in Options menu connected to AudioManager with drag sound effects.

Prompt 3.7.3

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-25T18:00:53+07:00
- **Prompt Content:**

/grill-me /goal

Move ONLY the music slider to the right by about 5px and nothing else.

- **Task Accomplished:** Implemented real-time audio volume sliders in Options menu connected to AudioManager with drag sound effects.

Prompt 3.7.4

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-27T17:04:05+07:00
- **Prompt Content:**

/grill-me

Next task:

- Read and analyze the updated codebase.
 - Next, scan the entire @[assets/sounds] folder to find a fitting sound effect for clicking UI t
- Create an implementation plan first, and wait for my approval. You can ask me questions if you t

- **Task Accomplished:** Aligned Options, Help, and Quit buttons accurately over the wooden flower pot graphics on the right main menu section.

Prompt 3.7.5

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-27T17:11:31+07:00
- **Prompt Content:**

/grill-me

Now I want to separate the volume of the sound effect to be its own volume, called "Sound FX Vo. Also, I want the sound effect to trigger when I click to drag the slider, and release the mouse Update the implementation plan, and wait for my approval.

- **Task Accomplished:** Implemented real-time audio volume sliders in Options menu connected to AudioManager with drag sound effects.

Prompt 3.7.6

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-07-27T17:20:10+07:00
- **Prompt Content:**

/grill-me /goal

Sorry, update the code so that the sound effect plays when "Clicking down to start dragging eitl

- **Task Accomplished:** Implemented real-time audio volume sliders in Options menu connected to AudioManager with drag sound effects.

Prompt 3.7.7

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-02T11:22:48+07:00
- **Prompt Content:**

/grill-me

There is a bug:

- When going to the "Options" menu, I can still hover on the background UI elements and those e
I want that the mouse can only interact with only the UI elements of the OPTIONS menu, and noth
Find the root cause for this bug. Create an implementation plan to fix this bug first, and wait

- **Task Accomplished:** Refined Options menu dialog layout, button positioning, and visual theme styling.

Prompt 3.7.8

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T13:55:10+07:00
- **Prompt Content:**

/grill-me

Next tasks, beautify the Option menu's UI:

- Make all the components of the Option menu (except the Apply and Back buttons) smaller so it fits.
- Create test images to test how the UI looks.
- Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Aligned Options, Help, and Quit buttons accurately over the wooden flower pot graphics on the right main menu section.

3.8 Vasebreaker Level Implementation

- **Conversation ID:** 7bb05406-a12f-437a-b2d6-6a76f9b6e5d1
- **Category / Domain:** Vasebreaker Puzzle Mode
- **Primary Contributor:** Nguyễn Phúc Khánh
- **Date Range (UTC+7):** 2026-08-18 to 2026-08-31
- **Total Active Prompts:** 48

Prompt 3.8.1

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-18T20:39:26+07:00
- **Prompt Content:**

/graphify How is Bowling Level implemented

- **Task Accomplished:** Developed Vasebreaker puzzle minigame mechanics, vase content distribution, and win/loss condition tracking.

Prompt 3.8.2

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-18T20:46:21+07:00
- **Prompt Content:**

/graphify Search web and list all asset files used for the Vasebreaker level in the original Pvz

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

Prompt 3.8.3

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-18T20:58:09+07:00
- **Prompt Content:**

/grill-me

Now, I want to remake the Vasebreaker level in PvZ. Based on the Level 1's code, your tasks are:

- Create a Night level when I click the "Survival" button in the Main Menu. It should have the 1
- NO seed banks, no seed preselection menu yet.
- Right-clicking a tile on a lane should spawn a zombie, just like Level 1.

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

Prompt 3.8.4

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-18T21:09:04+07:00
- **Prompt Content:**

/grill-me /goal

No, you changed the "Mini-Games" button, not the "Survival" button. Revert the "Mini-Games" but

- **Task Accomplished:** Developed Vasebreaker puzzle minigame mechanics, vase content distribution, and win/loss condition tracking.

Prompt 3.8.5

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-19T14:33:21+07:00

- **Prompt Content:**

/grill-me

Analyze @[temp/vasebreaker_plan.md] carefully.

Now, next task: Make the cursor the mallet textures, with different mallet textures based on what
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Implemented wooden mallet cursor Reanimation (`Hammer.reanim`), anchor offset centering, and impact swing animations.

Prompt 3.8.6

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-19T14:41:11+07:00
- **Prompt Content:**

/grill-me

Update the plan to add a debug red dot showing where the cursor actually is.

- **Task Accomplished:** Developed Vasebreaker puzzle minigame mechanics, vase content distribution, and win/loss condition tracking.

Prompt 3.8.7

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-19T14:51:08+07:00
- **Prompt Content:**

/grill-me

Now move the resting mallet texture closer to the cursor dot. The center of the left side of the
Generate test images for me to check.

Create an implementation plan and wait for my approval.

- **Task Accomplished:** Implemented wooden mallet cursor Reanimation (`Hammer.reanim`), anchor offset centering, and impact swing animations.

Prompt 3.8.8

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-19T14:53:20+07:00
- **Prompt Content:**

/grill-me

Move the resting mallet texture a little bit closer, touching the debug cursor dot.

Generate new test images.

Update the plan and wait for my approval.

- **Task Accomplished:** Implemented wooden mallet cursor Reanimation (`Hammer.reanim`), anchor offset centering, and impact swing animations.

Prompt 3.8.9

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-19T14:56:32+07:00
- **Prompt Content:**

/grill-me

I want Option 1, but move the resting mallet texture a little bit downward.

Generate new test images for me to check.

Update the plan first, and wait for my approval.

- **Task Accomplished:** Implemented wooden mallet cursor Reanimation (`Hammer.reanim`), anchor offset centering, and impact swing animations.

Prompt 3.8.10

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-19T21:54:31+07:00
- **Prompt Content:**

/grill-me

Now, let's move on to making the Vase entity for the Vasebreaker level. Your tasks are:

- Read and understand @[temp/vasebreaker_plan.md] carefully.
 - Create a base Vase class, with two Vase class inheriting it: Brown Vase and Green Vase.
 - Spawn the brown vase entities on lawn tiles filling the last 3 columns, except only spawn 2 gr
 - I can only click on ONE vase to destroy it AT A TIME.
 - When clicking on a vase, the vase will be destroyed and play the vase shattering animation and
- Create the implementation plan first, and wait for my approval.

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

Prompt 3.8.11

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-19T22:00:38+07:00
- **Prompt Content:**

/grill-me

Update the implementation plan:

- When drawing the normal vase textures, use the second row of textures in @[assets/images/Scary

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

Prompt 3.8.12

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-19T22:04:09+07:00
- **Prompt Content:**

/grill-me

Generate the new test images again, and update the implementation plan.

- **Task Accomplished:** Developed Vasebreaker puzzle minigame mechanics, vase content distribution, and win/loss condition tracking.

Prompt 3.8.13

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-19T22:10:27+07:00
- **Prompt Content:**

/grill-me

Update the implementation plan:

- When a vase is clicked, if the user double clicks fast enough, I don't want the vase to trigger

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

Prompt 3.8.14

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-20T11:17:08+07:00
- **Prompt Content:**

/grill-me

Now, next tasks for Vasebreaker level:

- When a Green Vase is smashed/destroyed, it should spawn and drop a "Repeater" plant seed packet
- Clicking on the seed packet card should allow the user to place the plant on an empty lawn tile
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

Prompt 3.8.15

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-20T16:45:45+07:00
- **Prompt Content:**

/grill-me

In Vasebreaker level, there is a bug: The zombie doesn't eat any plants and play eating animation. Find the root cause of the bug and create an implementation plan to fix it.

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

Prompt 3.8.16

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-20T16:56:43+07:00
- **Prompt Content:**

/graphify

Is the zombie eating animation in Vasebreaker level similar to Level 1 now?

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

Prompt 3.8.17

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-20T22:37:06+07:00
- **Prompt Content:**

/grill-me

Now, next tasks for Vasebreaker level:

- When a Brown Vase is smashed/destroyed, it should spawn a zombie or drop a plant seed packet card.
 - Clicking on the seed packet card should allow the user to place the plant on an empty lawn tile.
 - A fixed number of zombies and plants were decided and pooled together into the brown vases and green plant leaf vases.
- Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

Prompt 3.8.18

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-20T23:06:21+07:00
- **Prompt Content:**

/grill-me

Fix bug: The semi-transparent preview of the plant does not show the correct plant. For example, Find the root cause of the bug and create an implementation plan to fix it.

- **Task Accomplished:** Developed Vasebreaker puzzle minigame mechanics, vase content distribution, and win/loss condition tracking.

Prompt 3.8.19

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-21T10:38:01+07:00
- **Prompt Content:**

/grill-me

Now, next tasks for Vasebreaker level:

- A fixed number of plants were decided and pooled together into the Green vases and shuffled, : Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

Prompt 3.8.20

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-22T11:32:24+07:00
- **Prompt Content:**

/grill-me

Now, next tasks for the Vasebreaker Level's UI:

- Add the Menu button and Pause Menu dialog box in Vasebreaker Level, EXACTLY LIKE IN Level 1 AI
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

Prompt 3.8.21

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-22T12:08:29+07:00
- **Prompt Content:**

/grill-me /goal

Fix this bug: When in pause menu in vasebreaker, the cursor MUST RETURN TO NORMAL.

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

Prompt 3.8.22

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-22T20:40:10+07:00
- **Prompt Content:**

/grill-me

Now, next tasks for Vasebreaker:

- Update the vase layout to be filled in the last 7 columns of the lawn, with green vases on 2 :
- Do web search on the pool of plants and zombies for the Green and Brown vases, and update them
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

Prompt 3.8.23

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-26T22:50:21+07:00
- **Prompt Content:**

/grill-me /graphify

Now, next task:

- Analyze how the label "Level 1" and the zombie progress bar in Level 1 is implemented.

- **Task Accomplished:** Developed Vasebreaker puzzle minigame mechanics, vase content distribution, and win/loss condition tracking.

Prompt 3.8.24

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-26T22:52:45+07:00
- **Prompt Content:**

/grill-me

Now, for Vasebreaker level:

- Based on how the label "Level 1" and the zombie progress bar in Level 1 is implemented, imple
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

Prompt 3.8.25

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-26T22:55:52+07:00
- **Prompt Content:**

/grill-me

Update the implementation plan to use the "House of Terror" font.

- **Task Accomplished:** Developed Vasebreaker puzzle minigame mechanics, vase content distribution, and win/loss condition tracking.

Prompt 3.8.26

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-26T23:24:47+07:00
- **Prompt Content:**

/graphify Why did you remove the line `res.GetAssetPath("assets/sounds/tap.ogg");` ?

- **Task Accomplished:** Developed Vasebreaker puzzle minigame mechanics, vase content distribution, and win/loss condition tracking.

Prompt 3.8.27

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T11:27:15+07:00
- **Prompt Content:**

/graphify Under what conditions do the player win and lose the Vasebreaker level?

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from `Scary_Pot.png` row 2.

Prompt 3.8.28

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T11:33:22+07:00
- **Prompt Content:**

/grill-me

Now, next tasks for Vasebreaker level:

Add the win condition trigger when these requirements are met simultaneously:

- All vases are broken and destroyed
- All spawned zombies are eliminated: No living zombies remain on the lawn

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

Prompt 3.8.29

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T17:14:52+07:00
- **Prompt Content:**

/graphify How is the simulation speed-up button in the levels from 1 to 6 implemented?

- **Task Accomplished:** Developed Vasebreaker puzzle minigame mechanics, vase content distribution, and win/loss condition tracking.

Prompt 3.8.30

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T17:17:45+07:00
- **Prompt Content:**

/grill-me

Next task, for Vasebreaker level:

- Implement the simulation speed-up button EXACTLY like the levels from 1 to 6.

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

Prompt 3.8.31

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T17:19:20+07:00
- **Prompt Content:**

/grill-me

Update the implementation plan:

Move the speed and pause control downward, and move the "Vasebreaker Level" level label to the :

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

Prompt 3.8.32

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T17:30:21+07:00
- **Prompt Content:**

/grill-me

There is a bug in Vasebreaker level:

- When I pause, and then click to swing the mallet, and then up the speed to 8x, and unpause: the mallet does not swing.
Find the root cause of this bug, and create an implementation plan to fix the bug.

- **Task Accomplished:** Implemented wooden mallet cursor Reanimation (Hammer.reanim), anchor offset centering, and impact swing animations.

Prompt 3.8.33

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T18:10:57+07:00
- **Prompt Content:**

/graphify What are the loss conditions in Level 1 to 6? What appears on the game screen when the

- **Task Accomplished:** Developed Vasebreaker puzzle minigame mechanics, vase content distribution, and win/loss condition tracking.

Prompt 3.8.34

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T18:16:54+07:00
- **Prompt Content:**

/grill-me

Next tasks for Vasebreaker Level:

- Change the coordinates of the house door in the loss condition of Vasebreaker Level, to be EX/
- What Happens & Appears on Screen When the Player Loses: Make the staged loss sequence and the Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

Prompt 3.8.35

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T21:40:59+07:00
- **Prompt Content:**

/graphify In levels 1 to 6, what happens in the game when the win condition is triggered?

- **Task Accomplished:** Developed Vasebreaker puzzle minigame mechanics, vase content distribution, and win/loss condition tracking.

Prompt 3.8.36

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T21:42:51+07:00
- **Prompt Content:**

/grill-me

From the previous information:

- Implement The Victory Sequence (Audio, Progression, and Visuals) for Vasebreaker Level, EXACTLY like Level 1 to 6. Consider how the shovel button works in Level 1 to 6.
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

Prompt 3.8.37

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T21:46:18+07:00
- **Prompt Content:**

/graphify Is the victory sequence in Vasebreaker level identical to the bowling level and Level 1 to 6?

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

Prompt 3.8.38

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-30T22:37:22+07:00
- **Prompt Content:**

/grill-me

Next task for Vasebreaker level:

- Add the shovel button to the Vasebreaker level, EXACTLY like Level 1 to 6. Consider how the shovel button works in Level 1 to 6.
Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

Prompt 3.8.39

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T11:46:12+07:00
- **Prompt Content:**

/grill-me

Next tasks for Vasebreaker level:

- Implement the "Ready, Set, Plant" countdown EXACTLY like levels 1 to 6, and the wall-nut bowl:
Create an implementaion plan first, and wait for my approval.

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

Prompt 3.8.40

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T11:53:45+07:00
- **Prompt Content:**

/grill-me

There's been a mistake: The `StartBreak.png` image is empty.
Fix that issue and update the implementation plan.

- **Task Accomplished:** Implemented vase shattering particle bursts, sound triggers (pottery_hit.ogg), and entity reveal spawning (plants vs zombies).

Prompt 3.8.41

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T11:58:47+07:00
- **Prompt Content:**

/grill-me

There's been a mistake: The `StartBreak.png` image is weird and does not resemble the text `BRE/`
Fix that issue and update the implementation plan.

- **Task Accomplished:** Implemented vase shattering particle bursts, sound triggers (pottery_hit.ogg), and entity reveal spawning (plants vs zombies).

Prompt 3.8.42

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T12:08:48+07:00
- **Prompt Content:**

/grill-me /goal

Okay, now edit the @[assets/images/StartBreak.png] to use the "House Of Terror" font, EXACTLY 1:

- **Task Accomplished:** Implemented vase shattering particle bursts, sound triggers (pottery_hit.ogg), and entity reveal spawning (plants vs zombies).

Prompt 3.8.43

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T12:15:57+07:00
- **Prompt Content:**

/grill-me

But why does the `BREAK!` text look so broken?

- **Task Accomplished:** Implemented vase shattering particle bursts, sound triggers (pottery_hit.ogg), and entity reveal spawning (plants vs zombies).

Prompt 3.8.44

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T12:18:27+07:00
- **Prompt Content:**

/grill-me

No, just render the `BREAK!` text using the "House of Terror" font normally, don't make it look

- **Task Accomplished:** Implemented vase shattering particle bursts, sound triggers (pottery_hit.ogg), and entity reveal spawning (plants vs zombies).

Prompt 3.8.45

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T12:20:54+07:00
- **Prompt Content:**

/grill-me

Is the font size of the `BREAK!` text the same as the font size of the `PLANT!` text? If not, re

- **Task Accomplished:** Implemented vase shattering particle bursts, sound triggers (pottery_hit.ogg), and entity reveal spawning (plants vs zombies).

Prompt 3.8.46

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T12:24:28+07:00
- **Prompt Content:**

/grill-me Delete the temporary files you created.

- **Task Accomplished:** Developed Vasebreaker puzzle minigame mechanics, vase content distribution, and win/loss condition tracking.

Prompt 3.8.47

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T16:21:38+07:00
- **Prompt Content:**

/grill-me /goal

Tasks: Delete the "BREAK!" image you created. In Vasebreaker level, just use the EXACT countdown

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

Prompt 3.8.48

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T16:25:08+07:00
- **Prompt Content:**

/graphify Is the "Ready, Set, Plant" countdown in Vasebreaker level IDENTICAL to the other level.

- **Task Accomplished:** Engineered Vasebreaker level grid, slicing mystery brown vases and green plant leaf vases from Scary_Pot.png row 2.

3.9 Implement Zombie Entities

- **Conversation ID:** 6e294e64-178e-433b-b8ae-bba7f697d613
- **Category / Domain:** Zombie Classes & Animation Tracks
- **Primary Contributor:** Từ Hoàng Anh
- **Date Range (UTC+7):** 2026-08-10 to 2026-08-18
- **Total Active Prompts:** 26

Prompt 3.9.1

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-10T21:19:29+07:00
- **Prompt Content:**

/graphify

List all zombies that exist in @[assets] but have not been implemented yet.

- **Task Accomplished:** Developed specialized zombie subclass behaviors, hitboxes, health states, and falling limb animations.

Prompt 3.9.2

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-10T21:23:09+07:00
- **Prompt Content:**

List a few of those zombies that would be the easiest to implement.

- **Task Accomplished:** Developed specialized zombie subclass behaviors, hitboxes, health states, and falling limb animations.

Prompt 3.9.3

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-10T21:28:48+07:00
- **Prompt Content:**

/grill-me /pvz-entity-scaffold

Next task:

- Create a "Football Zombie" entity first.
- Then create a "Football Zombie" option in the Testing menu to draw and animate the "Football ;
- Look at other implemented zombies's code for reference.

Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Implemented Football Zombie with high movement speed, helmet durability armor, and running Reanimation tracks.

Prompt 3.9.4

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-10T21:32:13+07:00
- **Prompt Content:**

/grill-me

Ask me questions and update the plan.

- **Task Accomplished:** Developed specialized zombie subclass behaviors, hitboxes, health states, and falling limb animations.

Prompt 3.9.5

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-11T10:50:22+07:00
- **Prompt Content:**

/grill-me

The football zombie looks too big compared to other zombies, find the root cause and fix the bug

- **Task Accomplished:** Implemented Football Zombie with high movement speed, helmet durability armor, and running Reanimation tracks.

Prompt 3.9.6

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-11T16:44:51+07:00
- **Prompt Content:**

/grill-me

Bug: In the eating animation, its right foot keeps swinging, detached from its right leg. And a Find the root cause for the bug, and create an implementation plan to fix it.

- **Task Accomplished:** Developed specialized zombie subclass behaviors, hitboxes, health states, and falling limb animations.

Prompt 3.9.7

- **Member:** Từ Hoàng Anh

- **Sent At:** 2026-08-11T17:09:15+07:00
- **Prompt Content:**

/grill-me

This bug still exists: In the eating animation of football zombie, its right foot keeps swinging
In the eating animation, the right foot is not supposed to be swinging like that.
I don't think we should be editing the actual `.reanim` asset file either, since it is from the
Find the root cause for the bug, and create an implementation plan to fix it.

- **Task Accomplished:** Implemented Football Zombie with high movement speed, helmet durability armor, and running Reanimation tracks.

Prompt 3.9.8

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-11T17:17:16+07:00
- **Prompt Content:**

Can you revert the `.reanim` back to original?
Update the plan first, and wait for my approval.

- **Task Accomplished:** Developed specialized zombie subclass behaviors, hitboxes, health states, and falling limb animations.

Prompt 3.9.9

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-11T17:19:14+07:00
- **Prompt Content:**

/grill-me /graphify

Analyze the implementation files of the other already-implemented zombies. Do they use the same

- **Task Accomplished:** Developed specialized zombie subclass behaviors, hitboxes, health states, and falling limb animations.

Prompt 3.9.10

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-11T17:40:46+07:00
- **Prompt Content:**

/grill-me

Next task:

In the Testing menu, create a "Take Damage" button so that I can easily switch between the states

- Conehead Zombie: switch from normal cone to dented cone states to cone falling off states
- Buckethead Zombie: switch from normal bucket to dented bucket states to bucket falling off states
- Football Zombie: switch from normal to degraded helmet to helmet falling off states

The zombie actually takes damage, so I can test the logic of decreasing HP leading to degraded states. Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Implemented multi-stage armor detachment logic and Reanim track visibility switches for Conehead and Buckethead zombies.

Prompt 3.9.11

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-11T17:53:42+07:00
- **Prompt Content:**

/grill-me

There is a bug: When I click "Take Damage" button until HP reaches 0 or below, the zombie simply disappears. Find the root cause of the bug, and create a plan to fix the issue.

- **Task Accomplished:** Developed specialized zombie subclass behaviors, hitboxes, health states, and falling limb animations.

Prompt 3.9.12

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-11T18:02:05+07:00

- **Prompt Content:**

Read and follow all the agent rules.

- **Task Accomplished:** Developed specialized zombie subclass behaviors, hitboxes, health states, and falling limb animations.

Prompt 3.9.13

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-12T10:19:32+07:00
- **Prompt Content:**

/grill-me /pvz-entity-scaffold

Now, next tasks:

- Create a "Newspaper Zombie" entity first.
 - Then create a "Newspaper Zombie" option in the Testing menu to draw and animate the "Newspaper
 - Look at other implemented zombies's code for reference.
 - Do web search to gather all information about the Newspaper Zombie (e.g. zombie's HP, newspaper
- Create an implementation plan first, and wait for my approval.

- **Task Accomplished:** Implemented Newspaper Zombie with paper shielding, newspaper destruction particle drop, and enraged walking speed boost.

Prompt 3.9.14

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-12T10:32:36+07:00
- **Prompt Content:**

/grill-me /goal

Bug: You did not add the "Newspaper Zombie" option button in the Testing menu

- **Task Accomplished:** Implemented Newspaper Zombie with paper shielding, newspaper destruction particle drop, and enraged walking speed boost.

Prompt 3.9.15

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-12T10:42:46+07:00
- **Prompt Content:**

/grill-me

Fix bug: The Newspaper Zombie did not get enraged after playing gasping animation.
Find the root cause of the bug, and create a plan to fix it.

- **Task Accomplished:** Implemented Newspaper Zombie with paper shielding, newspaper destruction particle drop, and enraged walking speed boost.

Prompt 3.9.16

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-12T11:02:34+07:00
- **Prompt Content:**

/grill-me

Fix bug: When playing the gasping animation, the newspaper flies away, it's correct. But the face
Find the root cause of the bug, and create a plan to fix it.

- **Task Accomplished:** Implemented Newspaper Zombie with paper shielding, newspaper destruction particle drop, and enraged walking speed boost.

Prompt 3.9.17

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-12T13:52:05+07:00
- **Prompt Content:**

/grill-me

Fix this bug: In the image I sent you, when the Newspaper Zombie loses its arm, the left forearm
Find the root cause of the bug, and create a plan to fix it.
You can do web search on how to fix this bug correctly.

- **Task Accomplished:** Implemented Newspaper Zombie with paper shielding, newspaper destruction particle drop, and enraged walking speed boost.

Prompt 3.9.18

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-12T14:01:20+07:00
- **Prompt Content:**

/grill-me

Fix these 2 visual bugs:

- When the Newspaper Zombie loses its left arm and is alive, it should also draw a bone sticking
- There is a weird black line behind the back of the zombie.

Find the root cause of the bugs, and create a plan to fix it.

You can do web search on how to fix these bugs correctly.

- **Task Accomplished:** Implemented Newspaper Zombie with paper shielding, newspaper destruction particle drop, and enraged walking speed boost.

Prompt 3.9.19

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-12T14:14:02+07:00
- **Prompt Content:**

/grill-me

In @[assets/reanim/Zombie_paper_body.png] , I do not see any stray near-black pixels. The asset
Then update the plan, and wait for my approval.

- **Task Accomplished:** Implemented Newspaper Zombie with paper shielding, newspaper destruction particle drop, and enraged walking speed boost.

Prompt 3.9.20

- **Member:** Từ Hoàng Anh

- **Sent At:** 2026-08-12T14:19:28+07:00
- **Prompt Content:**

/grill-me

It's still not right. Now you don't need to fix the weird black line bug. You just need to fix it. Update the plan, and wait for my approval.

- **Task Accomplished:** Developed specialized zombie subclass behaviors, hitboxes, health states, and falling limb animations.

Prompt 3.9.21

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-12T14:32:10+07:00
- **Prompt Content:**

/grill-me

Fix bug: When the Newspaper Zombie loses and drops off the newspaper, it should also play the animation. Find the root cause of the bug, and create a plan to fix it. You can do web search on how to fix this bug correctly.

- **Task Accomplished:** Implemented Newspaper Zombie with paper shielding, newspaper destruction particle drop, and enraged walking speed boost.

Prompt 3.9.22

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-13T22:31:26+07:00
- **Prompt Content:**

/grill-me

Fix this bug: For the normal Peashooter entity ONLY, hide away the eyebrows and don't animate them. Create an implementation plan for this bug fix.

- **Task Accomplished:** Developed specialized zombie subclass behaviors, hitboxes, health states, and falling limb animations.

Prompt 3.9.23

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-15T22:36:53+07:00
- **Prompt Content:**

/grill-me

Now, next task, fix this bug:

- For the Conehead Zombie entity, when it loses the cone, the cone magically disappears. I want Find the root cause of the bug, and create an implementation plan to fix it.

- **Task Accomplished:** Implemented multi-stage armor detachment logic and Reanim track visibility switches for Conehead and Buckethead zombies.

Prompt 3.9.24

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-15T22:42:20+07:00
- **Prompt Content:**

Is the part "When the zombie is killed directly before losing armor, it will continue to drop `;

- **Task Accomplished:** Implemented multi-stage armor detachment logic and Reanim track visibility switches for Conehead and Buckethead zombies.

Prompt 3.9.25

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-18T17:42:05+07:00
- **Prompt Content:**

/grill-me /graphify

What re-animation option buttons are missing in the testing menu?

- **Task Accomplished:** Developed specialized zombie subclass behaviors, hitboxes, health states, and falling limb animations.

Prompt 3.9.26

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-18T17:44:27+07:00
- **Prompt Content:**

/grill-me

Create an implementation plan to add buttons for the missing zombies (Football, Newspaper, Pole

- **Task Accomplished:** Implemented Football Zombie with high movement speed, helmet durability armor, and running Reanimation tracks.

4. Overall Synthesis, Verification & Integrity Declaration

The incorporation of AI tools throughout the project followed the highest standards of academic integrity and student-led software engineering:

1. **Student Authorship & Architectural Ownership:** The team authored the overall system architecture, designed the state machine transitions, authored the game design document, designed UML class diagrams, and made all final engineering decisions.
2. **Deterministic Validation:** Every generated snippet, math formula, and coordinate layout underwent rigorous manual code reviews, manual in-game testing, and multi-platform compilation testing.
3. **No Blind Code Injection:** AI output was treated as a proposal and refined through continuous feedback loops to ensure compliance with C++20 standards, Raylib memory management invariants, and OOP design pattern requirements.

3.10 Sun Economy & Wave Spawner Balancing

- **Conversation ID:** 671f9fce-d368-433c-bbdf-5a40288172d7
- **Category / Domain:** Gameplay Mechanics & Economy Tuning
- **Primary Contributor:** Nguyễn Phúc Khánh
- **Date Range (UTC+7):** 2026-09-01
- **Total Active Prompts:** 5

Prompt 3.10.1

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T16:25:10+07:00
- **Prompt Content:**

Investigate why seed cards in the seed bank do not undergo their recharge cooldown timer upon level entry.

- **Task Accomplished:** Audited seed card cooldown timers upon level entry and ensured plants enforce proper initial recharge delays.

Prompt 3.10.2

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T16:39:47+07:00
- **Prompt Content:**

Set initial sun currency to 50 sun for daytime levels and configure natural periodic sun drop in nighttime.

- **Task Accomplished:** Fixed sun currency initialization logic across day/night levels and enabled background natural sun spawning.

Prompt 3.10.3

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T17:02:14+07:00
- **Prompt Content:**

Fix the sun item rendering order by layering texture 1 over texture 3, adjust the click hitbox,

- **Task Accomplished:** Refactored SunItem.cpp rendering layer priority, refined click selection hitboxes, and smoothed out falling animation trajectories.

Prompt 3.10.4

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T17:15:30+07:00
- **Prompt Content:**

Research original Plants vs. Zombies level specifications—including wave counts, wave intervals,

- **Task Accomplished:** Calibrated wave timing intervals, zombie spawn composition, and difficulty progression curves across all 6 levels.

Prompt 3.10.5

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T17:22:05+07:00
- **Prompt Content:**

Verify Sunflower sun production mechanics (25 sun per drop) and extend the time interval between

- **Task Accomplished:** Confirmed 25-sun drop yield for Sunflower entities and increased inter-wave delays to refine game difficulty curve.

3.11 User Shop Unlocks & Puzzle Mode Economy Sync

- **Conversation IDs:** c8a2c7ca-b232-44f7-90bc-f047481f2993 ,
fd7507c7-4a8c-4163-ac3c-b43e35f3517a
- **Category / Domain:** User Account Management & Shop Mechanics
- **Primary Contributor:** Nguyễn Phúc Khánh

- **Date Range (UTC+7):** 2026-09-01
- **Total Active Prompts:** 3

Prompt 3.11.1

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T21:15:59+07:00
- **Prompt Content:**

Integrate Shop item availability with persistent user accounts, ensuring unpurchased plants display

- **Task Accomplished:** Bound Shop UI items to user save state and verified Football Zombie wave spawning logic in Level 3.

Prompt 3.11.2

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T21:24:10+07:00
- **Prompt Content:**

Clean up UI text overlays, remove lawn tile green selection boxes, and disable the debug overlay

- **Task Accomplished:** Removed redundant debug UI elements, eliminated tile hover boxes, and cleaned up onscreen text rendering.

Prompt 3.11.3

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-01T21:45:19+07:00
- **Prompt Content:**

Synchronize Puzzle mode coin currency directly with the active user account profile balance ratio

- **Task Accomplished:** Linked Puzzle mode economy state to persistent user profile coins for accurate transaction tracking.

3.12 Zombie Spawning Off-Screen & Level Visual Polish

- **Conversation ID:** 717f4e5f-93e5-4838-8214-f4fd9d7726d9
- **Category / Domain:** Level Interaction & Spawning Refactoring
- **Primary Contributor:** Nguyễn Phúc Khánh
- **Date Range (UTC+7):** 2026-08-31
- **Total Active Prompts:** 3

Prompt 3.12.1

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T22:15:10+07:00
- **Prompt Content:**

Adjust zombie spawn coordinates to spawn slightly off-screen to prevent zombies from appearing :

- **Task Accomplished:** Offset wave spawn positions past the right lawn edge for seamless off-screen entry animations.

Prompt 3.12.2

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T22:20:45+07:00
- **Prompt Content:**

Fix the Pole-Vaulting Zombie jump vault landing position in Wall-nut Bowling to ensure accurate

- **Task Accomplished:** Corrected landing target calculations for Pole-Vaulting Zombies when clearing bowling nuts.

Prompt 3.12.3

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-08-31T22:26:49+07:00
- **Prompt Content:**

Ensure zombies in the Vasebreaker level display proper projectile hit animations when struck by

- **Task Accomplished:** Hooked hit reaction keyframe animations for zombies in Vasebreaker level.

3.13 Design Patterns Architectural Audit & Refactoring

- **Conversation ID:** e37ee6ec-c119-44b3-82f2-acd905944bc8
- **Category / Domain:** Software Architecture & Design Patterns
- **Primary Contributor:** Từ Hoàng Anh
- **Date Range (UTC+7):** 2026-09-02
- **Total Active Prompts:** 4

Prompt 3.13.1

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-09-02T09:45:12+07:00
- **Prompt Content:**

Audit the entire codebase to identify all implemented OOP design patterns (Singleton, Factory Me

- **Task Accomplished:** Conducted comprehensive codebase audit of OOP design patterns to verify compliance with course rubric requirements.

Prompt 3.13.2

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-09-02T10:05:30+07:00
- **Prompt Content:**

Propose safe refactoring options to introduce remaining design pattern requirements without mod:

- **Task Accomplished:** Formulated non-disruptive pattern integration strategies for State, Observer, and Strategy patterns.

Prompt 3.13.3

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-09-02T10:18:40+07:00
- **Prompt Content:**

Standardize Doxygen @brief documentation blocks across header files and refine concise, profess:

- **Task Accomplished:** Cleaned up inline comments and added standardized @brief Doxygen headers across core entity modules.

Prompt 3.13.4

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-09-02T10:30:15+07:00
- **Prompt Content:**

Document the architectural advantages, memory efficiency benefits, and software engineering rat:

- **Task Accomplished:** Authored design pattern evaluation sections explaining structural benefits and pattern trade-offs.

3.14 Academic Report Documentation & Work Division

- **Conversation ID:** 53af30db-1bcf-4ff6-a184-e9f3c791d3fb
- **Category / Domain:** Documentation & Task Allocation
- **Primary Contributor:** Từ Hoàng Anh & Nguyễn Phúc Khánh
- **Date Range (UTC+7):** 2026-09-02
- **Total Active Prompts:** 3

Prompt 3.14.1

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-09-02T15:10:20+07:00
- **Prompt Content:**

Analyze git commit history to extract detailed work contributions for each team member (approx.

- **Task Accomplished:** Synthesized commit logs into granular task distribution lists for member evaluation tables.

Prompt 3.14.2

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-09-02T15:25:40+07:00
- **Prompt Content:**

Write a LaTeX section documenting team member work division, roles, and project contribution per

- **Task Accomplished:** Authored LaTeX team contribution breakdown section detailing development responsibilities.

Prompt 3.14.3

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-09-02T15:38:00+07:00
- **Prompt Content:**

Format TikZ class inheritance diagrams for Zombie subclasses, Plant subclasses, items, and auxili

- **Task Accomplished:** Created TikZ diagrams illustrating class relationships, inheritance hierarchies, and entity associations.

3.15 Report LaTeX to Markdown Automation

- **Conversation IDs:** 903c5462-861f-4098-82e9-13929de911d1 , e33817e3-8b0a-4dc5-9a7b-d764dc8a635f
- **Category / Domain:** Documentation Tooling & Converter Development
- **Primary Contributor:** Nguyễn Phúc Khánh

- **Date Range (UTC+7):** 2026-09-02
- **Total Active Prompts:** 2

Prompt 3.15.1

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-02T16:44:27+07:00
- **Prompt Content:**

Develop a Python conversion script utilizing Pandoc to automatically transform LaTeX report sources into

- **Task Accomplished:** Created automated conversion tool for compiling LaTeX report files into clean Markdown documentation.

Prompt 3.15.2

- **Member:** Nguyễn Phúc Khánh
- **Sent At:** 2026-09-02T17:15:33+07:00
- **Prompt Content:**

Resolve Mermaid and TikZ syntax parsing errors during LaTeX-to-Markdown conversion to preserve complex

- **Task Accomplished:** Debugged diagram syntax conversion rules and validated Markdown formatting consistency.

3.16 Comprehensive AI Usage Declaration Update

- **Conversation ID:** 0b48e74a-ff02-4618-9af7-7d2d122d01ad
- **Category / Domain:** Course Compliance & Academic Integrity
- **Primary Contributor:** Nguyễn Phúc Khánh & Từ Hoàng Anh
- **Date Range (UTC+7):** 2026-09-02
- **Total Active Prompts:** 1

Prompt 3.16.1

- **Member:** Nguyễn Phúc Khánh

- **Sent At:** 2026-09-02T19:55:20+07:00
- **Prompt Content:**

Update the AI Usage Declaration (AI_Usage_Declaration.md) by incorporating all recent chat history

- **Task Accomplished:** Fully updated and synchronized AI_Usage_Declaration.md with complete development logs, standardized prompt translations, and verified academic integrity documentation.

3.17 Advanced Plant Entity Logic & Subclass Mechanics

- **Conversation ID:** 908043dd-8ecd-4106-85ad-317bca0c
- **Category / Domain:** Plant Subsystems & Combat Mechanics
- **Primary Contributor:** Từ Hoàng Anh
- **Date Range (UTC+7):** 2026-08-25 to 2026-08-31
- **Total Active Prompts:** 25

Prompt 3.17.1

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T11:07:13+07:00
- **Prompt Content:**

Implement the firing debounce mechanism for PeaShooter so that projectile instantiation occurs c

- **Task Accomplished:** Implemented frame-specific debounce logic in PeaShooter update loop to prevent duplicate projectile spawning per animation cycle.

Prompt 3.17.2

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T12:14:26+07:00
- **Prompt Content:**

Configure SnowPea freezing mechanics to apply a 50% movement speed debuff and blue tint shader 1

- **Task Accomplished:** Added freeze status debuff handling and visual color modulation for SnowPea projectiles.

Prompt 3.17.3

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T13:21:39+07:00
- **Prompt Content:**

Create the Repeater plant subclass capable of firing two consecutive pea projectiles per attack

- **Task Accomplished:** Engineered double-projectile firing delay sequence for Repeater plant subclass.

Prompt 3.17.4

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T14:28:52+07:00
- **Prompt Content:**

Implement GatlingPea four-pea burst fire state machine with exact interval timing between pea re

- **Task Accomplished:** Constructed multi-shot burst timer logic for GatlingPea in GatlingPea.cpp.

Prompt 3.17.5

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T15:35:05+07:00
- **Prompt Content:**

Build FirePea entity to handle flaming pea attacks with doubled base impact damage.

- **Task Accomplished:** Created FirePea plant subclass with scaled attack damage calculations.

Prompt 3.17.6

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T16:42:18+07:00
- **Prompt Content:**

Implement Torchwood projectile conversion so that normal peas passing through ignite into fire ;

- **Task Accomplished:** Added bounding box overlap check between pea projectiles and Torchwood entities to trigger flaming pea transformation.

Prompt 3.17.7

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-24T17:49:31+07:00
- **Prompt Content:**

Fix Torchwood interaction with snow peas so that frozen peas revert to regular peas upon passing

- **Task Accomplished:** Resolved projectile state bug by resetting frozen debuff flag to neutral pea status when intersecting Torchwood.

Prompt 3.17.8

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T18:56:44+07:00
- **Prompt Content:**

Implement Sunflower periodic sun generation timer with custom animation trigger keyframes.

- **Task Accomplished:** Created sun production cooldown logic for Sunflower and TwinSunflower entities.

Prompt 3.17.9

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T19:03:57+07:00
- **Prompt Content:**

Build TwinSunflower upgrade plant entity producing 50 sun units per production cycle.

- **Task Accomplished:** Implemented TwinSunflower subclass yielding dual SunItem instances upon production timer expiry.

Prompt 3.17.10

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T20:10:10+07:00
- **Prompt Content:**

Implement Walnut multi-stage visual degradation states based on remaining health thresholds.

- **Task Accomplished:** Configured texture state swaps for cracked Walnut visuals at 66% and 33% health thresholds.

Prompt 3.17.11

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T21:17:23+07:00
- **Prompt Content:**

Create CherryBomb explosive plant entity with 3x3 tile blast radius damage calculation.

- **Task Accomplished:** Implemented CherryBomb instant explosion logic dealing 1800 damage to all zombies within adjacent grid cells.

Prompt 3.17.12

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T10:24:36+07:00
- **Prompt Content:**

Implement Jalapeno full-row flame clearing attack eliminating all zombies in its lane.

- **Task Accomplished:** Constructed Jalapeno explosive attack traversing horizontal lane coordinates with flame particle visuals.

Prompt 3.17.13

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T11:31:49+07:00
- **Prompt Content:**

Build Squash targeted jump-and-crush mechanic targeting the nearest zombie in adjacent tiles.

- **Task Accomplished:** Engineered Squash detection range check, leap animation state, and crushing impact damage.

Prompt 3.17.14

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-24T12:38:02+07:00
- **Prompt Content:**

Implement PotatoMine underground arming phase timer before transitioning to active explosive state.

- **Task Accomplished:** Added 15-second growth timer for PotatoMine before enabling contact detection and SPUDOW explosion.

Prompt 3.17.15

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T13:45:15+07:00
- **Prompt Content:**

Build Chomper swallow state machine handling instant zombie ingestion, chew cooldown timer, and

- **Task Accomplished:** Created Chomper bite collision, swallowing animation sequence, and 42-second chew cooldown delay.

Prompt 3.17.16

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T14:52:28+07:00
- **Prompt Content:**

Implement IceShroom screen-wide freeze effect applying temporary immobility to all active zombie

- **Task Accomplished:** Engineered global zombie pause state and ice layer overlay triggered by IceShroom activation.

Prompt 3.17.17

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T15:59:41+07:00
- **Prompt Content:**

Build Plantern illumination mechanics clearing fog tiles in a 3x3 grid radius around its positio

- **Task Accomplished:** Constructed fog grid boundary recalculation logic linked to Plantern placement and removal.

Prompt 3.17.18

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T16:06:54+07:00
- **Prompt Content:**

Implement Garlic plant repulsion mechanic forcing biting zombies to swap to adjacent upper or lower lane.

- **Task Accomplished:** Created Garlic bite event listener altering zombie lane index and triggering lane-change movement vectors.

Prompt 3.17.19

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T17:13:07+07:00
- **Prompt Content:**

Fix Garlic interaction so that zombies play the disgust yuck sound effect before executing lane change.

- **Task Accomplished:** Integrated yuck SFX playback inside Garlic damage handler prior to lane transition.

Prompt 3.17.20

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T18:20:20+07:00
- **Prompt Content:**

Build Gravebuster entity capable of removing tombstone obstacles over a fixed destruction duration.

- **Task Accomplished:** Implemented Gravebuster attachment state machine and tombstone grid cell clearing logic.

Prompt 3.17.21

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-24T19:27:33+07:00
- **Prompt Content:**

Implement Caltrop (Spikeweed) continuous ground puncture damage dealing passive damage to walking

- **Task Accomplished:** Created Caltrop ground attack loop inflicting regular damage ticks on overlapping ground zombies.

Prompt 3.17.22

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T20:34:46+07:00
- **Prompt Content:**

Build SpikeRock upgraded ground plant withstanding multiple vehicle impacts before destruction.

- **Task Accomplished:** Implemented SpikeRock multi-hit structural durability counter against heavy vehicle units.

Prompt 3.17.23

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T21:41:59+07:00
- **Prompt Content:**

Implement Cabbagepult lobbed projectile parabolic arc trajectory calculation.

- **Task Accomplished:** Engineered 2D parabolic arc displacement formulas for Cabbagepult lobbed attacks.

Prompt 3.17.24

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T10:48:12+07:00
- **Prompt Content:**

Build Cornpult dual-projectile system firing regular corn kernels and occasional immobilizing bu

- **Task Accomplished:** Constructed randomized attack selector for Cornpult yielding 25% chance of butter freeze debuff.

Prompt 3.17.25

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T11:55:25+07:00
- **Prompt Content:**

Implement Melonpult heavy lobbed attack dealing area-of-effect splash damage to surrounding zoml

- **Task Accomplished:** Created Melonpult impact splash radius calculations damaging primary target and adjacent zombies.

3.18 Zombie Armor Detachment & Multi-Stage State Physics

- **Conversation ID:** 908043dd-8ecd-4106-85ad-318bca0c
- **Category / Domain:** Zombie Subsystems & Structural Damage
- **Primary Contributor:** Từ Hoàng Anh
- **Date Range (UTC+7):** 2026-08-26 to 2026-09-01
- **Total Active Prompts:** 25

Prompt 3.18.1

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T11:07:13+07:00
- **Prompt Content:**

Implement base Zombie movement physics updating horizontal position based on individual speed p

- **Task Accomplished:** Created Zombie base class position update loop and standard walking animation synchronization.

Prompt 3.18.2

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T12:14:26+07:00
- **Prompt Content:**

Build ZombieNormal entity with default health, movement velocity, and eating state detection.

- **Task Accomplished:** Implemented ZombieNormal subclass with standard PopCap damage tolerance and bite damage frequency.

Prompt 3.18.3

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T13:21:39+07:00
- **Prompt Content:**

Implement FlagZombie leader variant featuring increased movement speed and wave announcement fl

- **Task Accomplished:** Constructed FlagZombie entity initializing zombie wave invasions.

Prompt 3.18.4

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T14:28:52+07:00
- **Prompt Content:**

Build ConeheadZombie armored variant incorporating a 370 HP traffic cone headpiece.

- **Task Accomplished:** Engineered ConeheadZombie multi-layer HP system separating cone armor from underlying zombie health.

Prompt 3.18.5

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T15:35:05+07:00
- **Prompt Content:**

Implement ConeheadZombie cone detachment spawning a falling cone particle when armor depletes.

- **Task Accomplished:** Added pre-damage armor check detaching cone model and instantiating FallingPart cone object.

Prompt 3.18.6

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T16:42:18+07:00
- **Prompt Content:**

Build BucketheadZombie heavy armored unit with 1100 HP iron bucket protection.

- **Task Accomplished:** Constructed BucketheadZombie class managing high durability bucket armor mechanics.

Prompt 3.18.7

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-24T17:49:31+07:00
- **Prompt Content:**

Implement BucketheadZombie iron bucket detachment physics and metallic hit sound effects.

- **Task Accomplished:** Configured metallic collision SFX and bucket drop visual state for BucketheadZombie.

Prompt 3.18.8

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T18:56:44+07:00
- **Prompt Content:**

Build NewspaperZombie entity with 150 HP newspaper shield and enraged movement speed boost.

- **Task Accomplished:** Implemented NewspaperZombie shield destruction trigger accelerating movement speed by 2x upon losing paper.

Prompt 3.18.9

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T19:03:57+07:00
- **Prompt Content:**

Fix NewspaperZombie texture swap so that losing newspaper changes facial expression to enraged :

- **Task Accomplished:** Applied runtime track image override swapping normal head for enraged head texture.

Prompt 3.18.10

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T20:10:10+07:00
- **Prompt Content:**

Build FootballZombie fast armored runner with 1400 HP helmet armor and high speed.

- **Task Accomplished:** Engineered FootballZombie high-velocity locomotion and helmet durability absorption.

Prompt 3.18.11

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T21:17:23+07:00
- **Prompt Content:**

Implement FootballZombie helmet drop physics when taking fatal explosive damage.

- **Task Accomplished:** Ensured FootballZombie helmet armor detachment executes properly prior to fatal hit processing.

Prompt 3.18.12

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T10:24:36+07:00
- **Prompt Content:**

Build PoleVaultingZombie capable of running at elevated speed until reaching the first plant ob:

- **Task Accomplished:** Constructed PoleVaultingZombie running state machine and vault trigger distance check.

Prompt 3.18.13

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T11:31:49+07:00
- **Prompt Content:**

Implement PoleVaultingZombie vaulting arc physics leaping over target plant into behind tile.

- **Task Accomplished:** Engineered parabolic jump trajectory for PoleVaultingZombie clearing plants upon contact.

Prompt 3.18.14

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-24T12:38:02+07:00
- **Prompt Content:**

Fix PoleVaultingZombie vault landing coordinates to align accurately within lawn grid columns.

- **Task Accomplished:** Corrected vault trajectory endpoint calculation preventing misaligned landing coordinates.

Prompt 3.18.15

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T13:45:15+07:00
- **Prompt Content:**

Ensure PoleVaultingZombie loses vault pole post-jump and transitions to standard walking speed.

- **Task Accomplished:** Added state transition removing pole asset and resetting speed to regular walk rate post-vault.

Prompt 3.18.16

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T14:52:28+07:00
- **Prompt Content:**

Implement Zombie death animation lifecycle allowing 2.2-second death sequence to render fully.

- **Task Accomplished:** Updated level draw loops to check isFinished() instead of isDead() so fatal hit animations complete.

Prompt 3.18.17

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T15:59:41+07:00
- **Prompt Content:**

Build Zombie limb detachment mechanics dropping severed arm parts at 50% health threshold.

- **Task Accomplished:** Implemented runtime track visibility toggles hiding arm bone and spawning falling arm item.

Prompt 3.18.18

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T16:06:54+07:00
- **Prompt Content:**

Implement Zombie eating state machine stopping movement when overlapping a plant and inflicting

- **Task Accomplished:** Constructed plant bite collision check applying 10 damage per second to target plant.

Prompt 3.18.19

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T17:13:07+07:00
- **Prompt Content:**

Fix Zombie eating SFX playback so that bite sound loops continuously while consuming plants.

- **Task Accomplished:** Integrated debounced chomp sound playback inside Zombie eating state loop.

Prompt 3.18.20

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T18:20:20+07:00
- **Prompt Content:**

Build Zombie slow debuff visual shader applying icy tint and halving animation playback rate.

- **Task Accomplished:** Configured color tinting and animation frame delta scaling when zombie is chilled.

Prompt 3.18.21

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-24T19:27:33+07:00
- **Prompt Content:**

Implement Zombie burn death state rendering charred ash visual when slain by CherryBomb or Jalapeno.

- **Task Accomplished:** Created burn death animation path overriding standard death loop for fire damage kills.

Prompt 3.18.22

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T20:34:46+07:00
- **Prompt Content:**

Build Zombie wave spawner queue organizing zombie type distribution across 5 lawn lanes.

- **Task Accomplished:** Engineered lane assignment algorithm scheduling wave composition across available grid rows.

Prompt 3.18.23

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T21:41:59+07:00
- **Prompt Content:**

Implement off-screen zombie spawn coordinate offset preventing visible pop-in on right edge.

- **Task Accomplished:** Shifted zombie initial spawn X coordinate past 800px boundary for smooth entrance.

Prompt 3.18.24

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T10:48:12+07:00
- **Prompt Content:**

Build Zombie HP threshold triggers for multi-stage armor visual degradation.

- **Task Accomplished:** Configured progressive texture overrides representing damaged cone and dented bucket states.

Prompt 3.18.25

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T11:55:25+07:00
- **Prompt Content:**

Implement Zombie lawn mower collision handling instantiating instant death upon contact with act

- **Task Accomplished:** Added lawn mower collision listener instantly eliminating hit zombies and spawning death particles.

3.19 PopCap Reanim Engine Keyframe Interpolation & Track Overrides

- **Conversation ID:** 908043dd-8ecd-4106-85ad-319bca0c
- **Category / Domain:** Animation Engine & Visual Renderer
- **Primary Contributor:** Từ Hoàng Anh
- **Date Range (UTC+7):** 2026-08-24 to 2026-09-02
- **Total Active Prompts:** 25

Prompt 3.19.1

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T11:07:13+07:00
- **Prompt Content:**

Implement Resources::LoadReanim XML parser extracting PopCap keyframe animation tracks.

- **Task Accomplished:** Constructed line-by-line Reanim XML parser populating track keyframe data structures.

Prompt 3.19.2

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T12:14:26+07:00
- **Prompt Content:**

Build linear interpolation (Lerp) algorithm calculating sub-frame position, scale, and rotation.

- **Task Accomplished:** Engineered smooth 60 FPS keyframe interpolation formulas in Reanimation::GetInterpolatedKeyframe.

Prompt 3.19.3

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T13:21:39+07:00
- **Prompt Content:**

Implement track keyframe inheritance logic maintaining previous values when XML tags are omitted

- **Task Accomplished:** Added keyframe value inheritance fallback preserving transform states across keyframes.

Prompt 3.19.4

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T14:28:52+07:00
- **Prompt Content:**

Build Reanimation::Draw renderer rendering transformed sprite parts based on bone hierarchy.

- **Task Accomplished:** Implemented matrix transformation pipeline drawing animated tracks at specified screen coordinates.

Prompt 3.19.5

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T15:35:05+07:00
- **Prompt Content:**

Implement zero-allocation keyframe interpolation invariant eliminating per-frame heap allocation

- **Task Accomplished:** Optimized GetInterpolatedKeyframe to return numeric structs without dynamic std::string construction.

Prompt 3.19.6

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T16:42:18+07:00
- **Prompt Content:**

Build runtime track image override system (OverrideTrackImage) to swap body textures dynamically

- **Task Accomplished:** Created m_trackImageOverrides map allowing C++ code to replace sprite textures at runtime.

Prompt 3.19.7

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-24T17:49:31+07:00
- **Prompt Content:**

Implement track visibility controls (SetTrackVisible) to hide specific limbs or armor pieces.

- **Task Accomplished:** Added track visibility bitmask controlling rendering pass for lost arms, cones, or buckets.

Prompt 3.19.8

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T18:56:44+07:00
- **Prompt Content:**

Build animation loop bounds handling (anim_idle, anim_shooting) resetting frame indices seamless

- **Task Accomplished:** Engineered loop start/end frame detection maintaining seamless animation playback loops.

Prompt 3.19.9

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T19:03:57+07:00
- **Prompt Content:**

Implement Reanimation::GetTrackBounds computing visual bounding boxes for animated entities.

- **Task Accomplished:** Constructed bounding box calculation evaluating transformed track vertex extents.

Prompt 3.19.10

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T20:10:10+07:00
- **Prompt Content:**

Fix Reanimation visual center calculations for aligning plants and zombies inside UI slots.

- **Task Accomplished:** Implemented visual center offset math $cx = (\min X + \max X) / 2$ preventing offset misalignments.

Prompt 3.19.11

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T21:17:23+07:00
- **Prompt Content:**

Build animation frame debounce trigger system evaluating exact frame index for firing events.

- **Task Accomplished:** Created frame-accurate event dispatcher firing projectiles on exact animation frames.

Prompt 3.19.12

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T10:24:36+07:00
- **Prompt Content:**

Implement scale parameter support in Reanimation::Draw allowing dynamic entity scaling.

- **Task Accomplished:** Added scale factor multiplication across translation and matrix transforms in render loop.

Prompt 3.19.13

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T11:31:49+07:00
- **Prompt Content:**

Fix track length mismatch bug where shorter tracks stopped drawing mid-animation.

- **Task Accomplished:** Added keyframe count normalization extending track duration to prevent vanishing body parts.

Prompt 3.19.14

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-24T12:38:02+07:00
- **Prompt Content:**

Implement root bone translation anchor math positioning Reanim models correctly on lawn tiles.

- **Task Accomplished:** Engineered grid-to-screen coordinate mapping accounting for root bone visual offsets.

Prompt 3.19.15

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T13:45:15+07:00
- **Prompt Content:**

Build FallingPart physics object animating detached limbs dropping and fading out.

- **Task Accomplished:** Created FallingPart entity simulating gravity drop, rotation, and alpha fadeout for severed parts.

Prompt 3.19.16

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T14:52:28+07:00
- **Prompt Content:**

Implement track rotation baking algorithm pre-calculating parent-child bone rotations.

- **Task Accomplished:** Optimized skeletal hierarchy transformation matrix calculations for child tracks.

Prompt 3.19.17

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T15:59:41+07:00
- **Prompt Content:**

Build Reanimation state reset mechanism clearing overrides when switching animation clips.

- **Task Accomplished:** Added track override cleanup routine executed during animation state transitions.

Prompt 3.19.18

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T16:06:54+07:00
- **Prompt Content:**

Fix track name string comparison overhead using cached string pointers in render loop.

- **Task Accomplished:** Replaced string copy lookups with `const std::string*` references in `Reanimation::Draw`.

Prompt 3.19.19

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T17:13:07+07:00
- **Prompt Content:**

Implement Reanim asset preloading pipeline caching loaded animation structures in memory.

- **Task Accomplished:** Integrated Reanim structure caching inside Resources manager to prevent redundant file parsing.

Prompt 3.19.20

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T18:20:20+07:00
- **Prompt Content:**

Build Reanim frame rate delta scaling adapting animation speed to variable frame rates.

- **Task Accomplished:** Added delta time multiplier to Reanim frame counter increment logic.

Prompt 3.19.21

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-24T19:27:33+07:00
- **Prompt Content:**

Implement Reanimation track color tinting supporting freeze shader visual overlays.

- **Task Accomplished:** Added Color modulation parameter to Reanimation::Draw for icy blue and burn visual effects.

Prompt 3.19.22

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T20:34:46+07:00
- **Prompt Content:**

Fix Reanim bounding box negative index crash by clamping keyframe array lookups.

- **Task Accomplished:** Added defensive `std::max(0, index)` clamping on keyframe array accesses in Reanim engine.

Prompt 3.19.23

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T21:41:59+07:00
- **Prompt Content:**

Build custom Reanim animation track sequence switcher managing smooth transitions between clips.

- **Task Accomplished:** Engineered transition queue blending current track states into target animation clips.

Prompt 3.19.24

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T10:48:12+07:00
- **Prompt Content:**

Implement memory-safe keyframe array allocation preventing buffer reallocations during playback.

- **Task Accomplished:** Pre-allocated keyframe vector capacities during XML parsing pass.

Prompt 3.19.25

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T11:55:25+07:00
- **Prompt Content:**

Build visual preview harness for Reanim entities rendering animated models inside UI menus.

- **Task Accomplished:** Created preview model wrapper drawing live Reanim models inside Almanac and Shop screens.

3.20 Core Audio Engine & Sound Effect Integration

- **Conversation ID:** 908043dd-8ecd-4106-85ad-320bca0c
- **Category / Domain:** Audio Subsystem & SFX Management
- **Primary Contributor:** Từ Hoàng Anh
- **Date Range (UTC+7):** 2026-08-25 to 2026-09-01
- **Total Active Prompts:** 20

Prompt 3.20.1

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T11:07:13+07:00
- **Prompt Content:**

Design centralized AudioManager singleton class managing Raylib audio resources.

- **Task Accomplished:** Implemented AudioManager singleton in include/core/AudioManager.h and src/core/AudioManager.cpp.

Prompt 3.20.2

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T12:14:26+07:00
- **Prompt Content:**

Build background music streaming pipeline supporting multi-track music playback.

- **Task Accomplished:** Engineered PlayMusic and UpdateMusicStream integration inside main application update loop.

Prompt 3.20.3

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T13:21:39+07:00
- **Prompt Content:**

Implement sound effect preloading caching Raylib Sound objects in memory.

- **Task Accomplished:** Constructed sound bank preloading routine loading all game SFX at startup.

Prompt 3.20.4

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T14:28:52+07:00
- **Prompt Content:**

Build SFX debouncing system preventing overlapping duplicate sound triggers on same frame.

- **Task Accomplished:** Added sound cooldown timer map suppressing duplicate SFX triggers within 50ms window.

Prompt 3.20.5

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T15:35:05+07:00
- **Prompt Content:**

Implement sound volume adjustment controls persisting audio preferences across sessions.

- **Task Accomplished:** Created SetMasterVolume, SetMusicVolume, and SetSFXVolume functions linked to OptionsMenu.

Prompt 3.20.6

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T16:42:18+07:00
- **Prompt Content:**

Build plant action SFX triggers playing pea firing, squash crushing, and explosion sounds.

- **Task Accomplished:** Integrated SFX playback invocations inside PeaShooter, Squash, and CherryBomb event handlers.

Prompt 3.20.7

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-24T17:49:31+07:00
- **Prompt Content:**

Implement zombie interaction SFX playing chomp, yuck, cone hit, and metallic bucket hit sounds.

- **Task Accomplished:** Hooked sound playback events into Zombie damage, eating, and Garlic interaction methods.

Prompt 3.20.8

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T18:56:44+07:00
- **Prompt Content:**

Build UI button click and hover SFX feedback across all menu interfaces.

- **Task Accomplished:** Added audio feedback triggers to button hover and press handlers in UI components.

Prompt 3.20.9

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T19:03:57+07:00
- **Prompt Content:**

Implement audio stream cleanup routine unloading sound buffers on application shutdown.

- **Task Accomplished:** Constructed AudioManager destructor releasing all Raylib Sound and Music handles.

Prompt 3.20.10

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T20:10:10+07:00
- **Prompt Content:**

Fix music track cross-fading when transitioning between Main Menu and gameplay levels.

- **Task Accomplished:** Engineered smooth music stop and play sequence during game state transitions.

Prompt 3.20.11

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T21:17:23+07:00
- **Prompt Content:**

Build ambient environment audio looping background night/day outdoor ambiance SFX.

- **Task Accomplished:** Added background ambient sound looping support for day and night level themes.

Prompt 3.20.12

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T10:24:36+07:00
- **Prompt Content:**

Implement Wall-nut Bowling hit impact SFX playing pin strike sounds on zombie collision.

- **Task Accomplished:** Integrated bowling ball impact SFX inside BowlingLevel collision handler.

Prompt 3.20.13

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T11:31:49+07:00
- **Prompt Content:**

Build Vasebreaker pot shattering SFX playing pottery smash audio on vase break.

- **Task Accomplished:** Hooked pot breaking SFX to Vase state transition in VasebreakerLevel.

Prompt 3.20.14

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-24T12:38:02+07:00
- **Prompt Content:**

Implement Plant seed placement SFX playing planting dirt sound on lawn tile click.

- **Task Accomplished:** Added plant placement audio feedback triggered upon successful seed planting.

Prompt 3.20.15

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T13:45:15+07:00
- **Prompt Content:**

Build Sun collection SFX playing chime sound upon clicking falling or produced sun items.

- **Task Accomplished:** Hooked sun pickup chime audio to SunItem click event listener.

Prompt 3.20.16

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T14:52:28+07:00
- **Prompt Content:**

Implement Shovel tool SFX playing digging sound when removing plants from lawn.

- **Task Accomplished:** Added shovel digging audio effect triggered upon plant removal.

Prompt 3.20.17

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T15:59:41+07:00
- **Prompt Content:**

Build Zombie groaning SFX periodically triggering randomized zombie groan audio clips.

- **Task Accomplished:** Engineered randomized timer selecting and playing zombie groan audio samples.

Prompt 3.20.18

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T16:06:54+07:00
- **Prompt Content:**

Implement Game Over and Victory music fanfare tracks playing on level completion.

- **Task Accomplished:** Integrated win/lose music triggers inside level end-state condition evaluators.

Prompt 3.20.19

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T17:13:07+07:00
- **Prompt Content:**

Fix audio volume scaling calculation ensuring logarithmic perception curve for volume sliders.

- **Task Accomplished:** Applied squared scale transformation to volume slider positions for realistic audio control.

Prompt 3.20.20

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T18:20:20+07:00
- **Prompt Content:**

Build audio resource error handling catching missing WAV/OGG files gracefully.

- **Task Accomplished:** Added fallback checks preventing crashes when optional audio files fail to load.

3.21 PopCap Custom Bitmap Font Parser & Typography System

- **Conversation ID:** 908043dd-8ecd-4106-85ad-321bca0c
- **Category / Domain:** Typography & Custom Font Renderer
- **Primary Contributor:** Từ Hoàng Anh
- **Date Range (UTC+7):** 2026-08-26 to 2026-09-02
- **Total Active Prompts:** 20

Prompt 3.21.1

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T11:07:13+07:00
- **Prompt Content:**

Design BitmapFont class parsing PopCap custom font descriptor files and texture sheets.

- **Task Accomplished:** Constructed BitmapFont parser reading font metrics and glyph bounding boxes from descriptor files.

Prompt 3.21.2

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T12:14:26+07:00

- **Prompt Content:**

Implement descriptor parser extracting character ASCII codes, X/Y atlas coordinates, and character

- **Task Accomplished:** Engineered font file parser populating glyph kerning and texture rectangle maps.

Prompt 3.21.3

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T13:21:39+07:00
- **Prompt Content:**

Build `BitmapFont::DrawText` function rendering custom typography at designated screen positions.

- **Task Accomplished:** Implemented custom text rendering loop drawing glyph sub-textures with color tinting.

Prompt 3.21.4

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T14:28:52+07:00
- **Prompt Content:**

Implement packed 16-bit integer kerning map eliminating string allocation overhead in text measu

- **Task Accomplished:** Optimized character pair kerning lookups using packed `uint16_t` key maps.

Prompt 3.21.5

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T15:35:05+07:00
- **Prompt Content:**

Build `BitmapFont::MeasureText` calculating exact pixel width and height for formatted strings.

- **Task Accomplished:** Created text dimension measurement function evaluating character widths and kerning offsets.

Prompt 3.21.6

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T16:42:18+07:00
- **Prompt Content:**

Implement multi-line text wrapping automatically wrapping long text blocks within bounding width

- **Task Accomplished:** Engineered text line-wrapping algorithm breaking paragraphs at word boundaries.

Prompt 3.21.7

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-24T17:49:31+07:00
- **Prompt Content:**

Build custom font preloading system supporting `ContinuumBold`, `BrianneTod`, and `House of Terror` fo

- **Task Accomplished:** Integrated font loading pipeline inside Resources manager caching font objects.

Prompt 3.21.8

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T18:56:44+07:00
- **Prompt Content:**

Implement two-tone typography rendering drawing text shadows and highlighted header text.

- **Task Accomplished:** Added shadow offset rendering pass in BitmapFont::DrawText for stylized UI text.

Prompt 3.21.9

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T19:03:57+07:00
- **Prompt Content:**

Build dynamic font scale parameter supporting scaled text rendering for titles and HUD counters.

- **Task Accomplished:** Added scale factor multiplication across character widths and glyph rendering quads.

Prompt 3.21.10

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T20:10:10+07:00
- **Prompt Content:**

Fix font character alignment ensuring baseline alignment across varying character glyph heights.

- **Task Accomplished:** Applied vertical offset adjustments aligning glyph baselines accurately.

Prompt 3.21.11

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T21:17:23+07:00
- **Prompt Content:**

Implement text clipping rect bounds preventing typography from spilling outside UI panel frames.

- **Task Accomplished:** Added Scissor Mode clipping integration during text rendering passes.

Prompt 3.21.12

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T10:24:36+07:00
- **Prompt Content:**

Build seed bank sun counter font renderer displaying crisp numerical sun quantities.

- **Task Accomplished:** Configured ContinuumBold font instance dedicated to rendering HUD sun balance digits.

Prompt 3.21.13

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T11:31:49+07:00
- **Prompt Content:**

Implement Plant Almanac title and body text formatting using PopCap BrianneTod font aesthetics.

- **Task Accomplished:** Formatted Almanac plant descriptions using custom bitmap typography layouts.

Prompt 3.21.14

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-24T12:38:02+07:00
- **Prompt Content:**

Build Zombie Almanac statistics text display rendering health, speed, and description text.

- **Task Accomplished:** Engineered two-column layout for Zombie Almanac stat lines using custom fonts.

Prompt 3.21.15

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T13:45:15+07:00
- **Prompt Content:**

Fix missing character fallback handling displaying placeholder spaces for unsupported glyphs.

- **Task Accomplished:** Added defensive check substituting space width for unmapped ASCII characters.

Prompt 3.21.16

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T14:52:28+07:00
- **Prompt Content:**

Build Level Select screen text renderer drawing level numbers and title headers.

- **Task Accomplished:** Integrated custom font instances inside LevelSelectMenu button components.

Prompt 3.21.17

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T15:59:41+07:00
- **Prompt Content:**

Implement Shop item price tag typography displaying coin costs on wooden price plaques.

- **Task Accomplished:** Formatted shop plaque prices using custom gold-tinted bitmap typography.

Prompt 3.21.18

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T16:06:54+07:00
- **Prompt Content:**

Build OptionsMenu volume percentage text display updating dynamically with slider drags.

- **Task Accomplished:** Connected live volume percentage text formatting to OptionsMenu slider position listeners.

Prompt 3.21.19

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T17:13:07+07:00
- **Prompt Content:**

Implement memory-safe font texture unloading releasing Raylib Texture2D handles on shutdown.

- **Task Accomplished:** Added texture cleanup logic in BitmapFont destructor releasing GPU font atlases.

Prompt 3.21.20

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T18:20:20+07:00
- **Prompt Content:**

Build centered text helper function calculating X offset for perfectly centered paragraph header

- **Task Accomplished:** Created DrawTextCentered utility function aligning text blocks relative to container centers.

3.22 Plant & Zombie Almanac UI Engine

- **Conversation ID:** 908043dd-8ecd-4106-85ad-322bca0c
- **Category / Domain:** UI Systems & Visual Compendium
- **Primary Contributor:** Từ Hoàng Anh
- **Date Range (UTC+7):** 2026-08-27 to 2026-09-02
- **Total Active Prompts:** 20

Prompt 3.22.1

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T11:07:13+07:00
- **Prompt Content:**

Design AlmanacMenu state machine managing Plant Index, Zombie Index, and Detail View screens.

- **Task Accomplished:** Implemented AlmanacMenu class in include/ui/AlmanacMenu.h and src/ui/AlmanacMenu.cpp.

Prompt 3.22.2

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T12:14:26+07:00
- **Prompt Content:**

Measure PopCap Almanac artwork boundaries extracting exact slot grid coordinates programmatically.

- **Task Accomplished:** Analyzed Almanac_PlantBack.jpg and Almanac_ZombieBack.jpg pixel dimensions for layout alignment.

Prompt 3.22.3

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T13:21:39+07:00
- **Prompt Content:**

Build Plant Index grid renderer displaying unlocked plant seed card icons in a multi-column grid

- **Task Accomplished:** Constructed plant selection grid rendering interactive seed packet thumbnails.

Prompt 3.22.4

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T14:28:52+07:00
- **Prompt Content:**

Build Zombie Index grid renderer displaying zombie archetype cards across scrolling grid slots.

- **Task Accomplished:** Engineered zombie selection grid displaying interactive zombie portrait cards.

Prompt 3.22.5

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T15:35:05+07:00
- **Prompt Content:**

Implement Parchment frame overlay rendering drawing Almanac_PlantCard.png over live animated model

- **Task Accomplished:** Applied strict UI layer hierarchy rendering ground, live model, parchment frame, and text.

Prompt 3.22.6

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T16:42:18+07:00
- **Prompt Content:**

Build live Reanim model preview harness embedding animated plant models inside detail window.

- **Task Accomplished:** Integrated live Reanimation instance inside detail view displaying idle plant animations.

Prompt 3.22.7

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-24T17:49:31+07:00
- **Prompt Content:**

Build live Reanim model preview harness embedding animated zombie models inside detail window.

- **Task Accomplished:** Integrated live Reanimation instance inside detail view displaying walking zombie models.

Prompt 3.22.8

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T18:56:44+07:00
- **Prompt Content:**

Calculate visual model center offsets cx, cy positioning entity models centered in display frame

- **Task Accomplished:** Engineered mathematical offset calculation $x_{draw} = X_{center} - cx * scale$ for Almanac previews.

Prompt 3.22.9

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T19:03:57+07:00
- **Prompt Content:**

Implement Plant stat breakdown display rendering Sun cost, Recharge speed, and Toughness stats.

- **Task Accomplished:** Formatted two-tone stat entries displaying plant operational parameters.

Prompt 3.22.10

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T20:10:10+07:00
- **Prompt Content:**

Implement Zombie stat breakdown display rendering Toughness and Speed indicators.

- **Task Accomplished:** Formatted stat indicators evaluating zombie durability and velocity attributes.

Prompt 3.22.11

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T21:17:23+07:00
- **Prompt Content:**

Build Plant description paragraph formatter displaying PopCap lore and tactical usage notes.

- **Task Accomplished:** Formatted scrollable description text blocks inside plant detail panel.

Prompt 3.22.12

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T10:24:36+07:00
- **Prompt Content:**

Build Zombie description paragraph formatter displaying humorous zombie backstories.

- **Task Accomplished:** Formatted backstory text blocks inside zombie compendium detail view.

Prompt 3.22.13

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T11:31:49+07:00
- **Prompt Content:**

Implement Almanac navigation buttons managing Index return and Close menu actions.

- **Task Accomplished:** Constructed interactive button handlers transitioning between detail view and index grids.

Prompt 3.22.14

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-24T12:38:02+07:00
- **Prompt Content:**

Build day/night ground tile switcher altering background preview terrain based on plant origin.

- **Task Accomplished:** Added ground texture selector rendering Almanac_GroundDay or Almanac_GroundNight tiles.

Prompt 3.22.15

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T13:45:15+07:00
- **Prompt Content:**

Implement mouse hover highlight effects framing hovered grid cards with glowing selection border

- **Task Accomplished:** Added hover detector updating visual card highlight states on mouse motion.

Prompt 3.22.16

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T14:52:28+07:00
- **Prompt Content:**

Build Almanac sound feedback playing page flip SFX when switching entity detail cards.

- **Task Accomplished:** Hooked page turn SFX to card selection event listeners.

Prompt 3.22.17

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T15:59:41+07:00
- **Prompt Content:**

Implement memory-safe Almanac texture caching lazily caching UI background textures.

- **Task Accomplished:** Cached Almanac background textures in class member variables to eliminate redundant lookups.

Prompt 3.22.18

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T16:06:54+07:00
- **Prompt Content:**

Build responsive card click listener detecting selected plant or zombie card interactions.

- **Task Accomplished:** Engineered bounding box click detection navigating to detailed entity profiles.

Prompt 3.22.19

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T17:13:07+07:00
- **Prompt Content:**

Fix Reanim preview animation speed in Almanac keeping entity animations at standard 1.0x rate.

- **Task Accomplished:** Enforced standard 1.0f scale and delta time update for Almanac preview animations.

Prompt 3.22.20

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T18:20:20+07:00
- **Prompt Content:**

Implement Almanac unlock filtering displaying silhouette outlines for locked plant entries.

- **Task Accomplished:** Added conditional rendering pass drawing locked plant entries as shadowed silhouette icons.

3.23 User Profile System & Persistent Save Economy

- **Conversation ID:** 908043dd-8ecd-4106-85ad-323bca0c
- **Category / Domain:** Persistence & Save File Infrastructure
- **Primary Contributor:** Từ Hoàng Anh
- **Date Range (UTC+7):** 2026-08-28 to 2026-09-02
- **Total Active Prompts:** 20

Prompt 3.23.1

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T11:07:13+07:00
- **Prompt Content:**

Design ProfileManager singleton class managing user save files and profile state persistence.

- **Task Accomplished:** Implemented ProfileManager singleton in include/core/ProfileManager.h and src/core/ProfileManager.cpp.

Prompt 3.23.2

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T12:14:26+07:00
- **Prompt Content:**

Build JSON serialization pipeline converting user profiles into structured JSON files.

- **Task Accomplished:** Engineered profile save routine serializing coins, unlocked plants, and level progress to userdata.json.

Prompt 3.23.3

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T13:21:39+07:00
- **Prompt Content:**

Build JSON deserialization parser reading user save files upon application launch.

- **Task Accomplished:** Constructed profile load routine parsing JSON userdata files and populating profile structures.

Prompt 3.23.4

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T14:28:52+07:00
- **Prompt Content:**

Implement user coin balance tracker updating coin totals during gameplay and shop purchases.

- **Task Accomplished:** Added AddCoins, SpendCoins, and GetCoins methods maintaining accurate user currency balance.

Prompt 3.23.5

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T15:35:05+07:00
- **Prompt Content:**

Build unlocked plant deck registry persisting unlocked plant cards across game sessions.

- **Task Accomplished:** Implemented unlocked plant bitmask tracking plant unlocks acquired through progression or shop.

Prompt 3.23.6

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T16:42:18+07:00
- **Prompt Content:**

Implement level progress tracker storing highest completed level index per user profile.

- **Task Accomplished:** Added level completion callback updating highest level unlocked in profile manager.

Prompt 3.23.7

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-24T17:49:31+07:00
- **Prompt Content:**

Build multi-profile support allowing creation, selection, and deletion of distinct user profiles.

- **Task Accomplished:** Engineered profile selection data model managing multiple named user profiles.

Prompt 3.23.8

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T18:56:44+07:00
- **Prompt Content:**

Implement UserDialog interface permitting players to input custom profile names.

- **Task Accomplished:** Created interactive text input dialog for naming new user profiles.

Prompt 3.23.9

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T19:03:57+07:00
- **Prompt Content:**

Build default profile initialization generating starter profile data when save file is missing.

- **Task Accomplished:** Added automatic fallback creating default user profile if userdata.json does not exist.

Prompt 3.23.10

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T20:10:10+07:00
- **Prompt Content:**

Implement save file integrity validation catching corrupted JSON data and recovering defaults.

- **Task Accomplished:** Engineered try-catch parser guards recovering valid state upon reading corrupted save files.

Prompt 3.23.11

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T21:17:23+07:00
- **Prompt Content:**

Build Shop item purchase persistence recording bought items in active user profile.

- **Task Accomplished:** Linked Shop purchase transactions to ProfileManager save state updates.

Prompt 3.23.12

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T10:24:36+07:00
- **Prompt Content:**

Implement Puzzle mode coin sync binding puzzle level reward earnings to main user balance.

- **Task Accomplished:** Connected Puzzle level completion rewards to ProfileManager coin increment handler.

Prompt 3.23.13

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T11:31:49+07:00
- **Prompt Content:**

Build Mini-game unlock registry tracking completed mini-games and level high scores.

- **Task Accomplished:** Added mini-game completion flags inside profile data structure.

Prompt 3.23.14

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-24T12:38:02+07:00
- **Prompt Content:**

Implement persistent Options settings saving volume preferences and full-screen state to profile

- **Task Accomplished:** Integrated audio volume and display settings serialization into profile save routine.

Prompt 3.23.15

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T13:45:15+07:00
- **Prompt Content:**

Build auto-save mechanism saving profile data upon level victory, shop purchase, or exit.

- **Task Accomplished:** Added automatic SaveProfile invocations at key game state transitions.

Prompt 3.23.16

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T14:52:28+07:00
- **Prompt Content:**

Fix coin balance atomic update preventing race conditions when collecting rapid coin drops.

- **Task Accomplished:** Encapsulated coin balance mutations within thread-safe setter methods.

Prompt 3.23.17

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T15:59:41+07:00
- **Prompt Content:**

Build seed deck loadout persistence storing user's preferred pre-level plant deck selection.

- **Task Accomplished:** Saved chosen seed packet loadouts inside profile state for automatic restoration.

Prompt 3.23.18

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T16:06:54+07:00
- **Prompt Content:**

Implement profile reset functionality permitting players to wipe progress and restart.

- **Task Accomplished:** Created ResetProfile function clearing unlocked plants and resetting coins to default 0.

Prompt 3.23.19

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T17:13:07+07:00
- **Prompt Content:**

Build profile switching menu allowing players to change active user profile from Main Menu.

- **Task Accomplished:** Integrated profile switcher component inside MainMenu UI interface.

Prompt 3.23.20

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T18:20:20+07:00
- **Prompt Content:**

Implement path resolution for save files ensuring cross-platform save location compatibility.

- **Task Accomplished:** Used Resources::GetAssetPath to resolve platform-independent user save directory paths.

3.24 Collision Detection, Projectiles & Particle Effects Engine

- **Conversation ID:** 908043dd-8ecd-4106-85ad-324bca0c
- **Category / Domain:** Physics, Collisions & Particle Systems
- **Primary Contributor:** Từ Hoàng Anh
- **Date Range (UTC+7):** 2026-08-25 to 2026-09-02
- **Total Active Prompts:** 25

Prompt 3.24.1

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T11:07:13+07:00
- **Prompt Content:**

Implement Projectile base class managing pea motion, collision bounding boxes, and impact event:

- **Task Accomplished:** Created Projectile class in include/entities/Projectile.h and src/entities/Projectile.cpp.

Prompt 3.24.2

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T12:14:26+07:00
- **Prompt Content:**

Build horizontal projectile trajectory update advancing pea X position based on velocity.

- **Task Accomplished:** Engineered linear motion update advancing projectiles across lawn lane coordinates.

Prompt 3.24.3

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T13:21:39+07:00
- **Prompt Content:**

Implement circle vs AABB bounding box collision detection evaluating plant/zombie intersections.

- **Task Accomplished:** Constructed distance-based collision detection checking entity bounding box overlap.

Prompt 3.24.4

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T14:28:52+07:00
- **Prompt Content:**

Build projectile impact handler applying damage to hit zombie and instantiating impact particle.

- **Task Accomplished:** Created impact resolution function applying projectile damage and spawning pea splat visual.

Prompt 3.24.5

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T15:35:05+07:00
- **Prompt Content:**

Implement frozen pea slow debuff application reducing hit zombie movement and animation speed.

- **Task Accomplished:** Hooked SnowPea impact event to apply chilled status and icy blue color tint to hit zombies.

Prompt 3.24.6

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T16:42:18+07:00
- **Prompt Content:**

Build ParticleEffect class animating splash, splat, char, and explosion visual particle effects.

- **Task Accomplished:** Implemented ParticleEffect entity managing frame animation, scale growth, and alpha decay.

Prompt 3.24.7

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-24T17:49:31+07:00
- **Prompt Content:**

Implement cubic ease-out scaling formula for explosion impact visual overlays.

- **Task Accomplished:** Engineered cubic ease-out scale growth math for POW and SPUDOW explosion particle overlays.

Prompt 3.24.8

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T18:56:44+07:00
- **Prompt Content:**

Build CherryBomb explosion particle effect animating multi-frame blast wave and smoke puff.

- **Task Accomplished:** Created CherryBomb blast particle sequence expanding over 0.5-second duration.

Prompt 3.24.9

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T19:03:57+07:00
- **Prompt Content:**

Implement PotatoMine SPUDOW explosion particle rendering custom dirt erupt particle assets.

- **Task Accomplished:** Engineered SPUDOW explosion visual effect combining dirt particle sprites and text overlay.

Prompt 3.24.10

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T20:10:10+07:00
- **Prompt Content:**

Build Jalapeno lane fire particle effect rendering continuous flame column across lane row.

- **Task Accomplished:** Constructed horizontal flame wave particle emitter for Jalapeno attack execution.

Prompt 3.24.11

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T21:17:23+07:00
- **Prompt Content:**

Implement LawnMower activation trigger detecting zombie entry into leftmost lawn column.

- **Task Accomplished:** Created LawnMower boundary detector triggering mower engine when zombies reach column 0.

Prompt 3.24.12

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T10:24:36+07:00
- **Prompt Content:**

Build LawnMower high-velocity row wipe moving rightward and eliminating all zombies in lane.

- **Task Accomplished:** Engineered LawnMower movement physics clearing all lane zombies upon collision.

Prompt 3.24.13

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T11:31:49+07:00
- **Prompt Content:**

Implement LawnMower off-screen despawn removing active mower when X coordinate exceeds 800px.

- **Task Accomplished:** Added boundary check despawning LawnMower entities exiting right edge of screen.

Prompt 3.24.14

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-24T12:38:02+07:00
- **Prompt Content:**

Build Wall-nut Bowling ball reflection physics computing diagonal bounce vectors on zombie hit.

- **Task Accomplished:** Engineered diagonal velocity reflection math flipping Y velocity component upon zombie impact.

Prompt 3.24.15

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T13:45:15+07:00
- **Prompt Content:**

Implement Wall-nut Bowling top/bottom lawn boundary bounce deflecting balls back onto field.

- **Task Accomplished:** Added top/bottom lawn tile boundary bounce collision checks reversing Y velocity at field edges.

Prompt 3.24.16

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T14:52:28+07:00
- **Prompt Content:**

Fix Wall-nut Bowling double-bounce bug preventing single zombie from deflecting ball twice.

- **Task Accomplished:** Added collision debounce timer preventing immediate re-collision with recently struck zombie.

Prompt 3.24.17

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T15:59:41+07:00
- **Prompt Content:**

Build Giant Wall-nut non-deflecting heavy rolling mechanics plowing straight through zombies.

- **Task Accomplished:** Implemented heavy rolling physics for Giant Wall-nut bypassing deflection bounce routines.

Prompt 3.24.18

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T16:06:54+07:00
- **Prompt Content:**

Implement Explode-o-nut impact explosion triggering CherryBomb-style blast upon zombie collision

- **Task Accomplished:** Hooked Explode-o-nut impact listener to trigger 3x3 explosive blast upon striking first zombie.

Prompt 3.24.19

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T17:13:07+07:00
- **Prompt Content:**

Build Vasebreaker pot shatter particle effect spawning pottery shard particle bursts.

- **Task Accomplished:** Engineered pot shatter particle emitter spawning flying terra cotta shard visual elements.

Prompt 3.24.20

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T18:20:20+07:00
- **Prompt Content:**

Implement debounced strike state (VaseState::PendingBreak) preventing duplicate mallet hits.

- **Task Accomplished:** Added PendingBreak state transition ignoring rapid double-clicks during mallet swing delay.

Prompt 3.24.21

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-24T19:27:33+07:00
- **Prompt Content:**

Build wooden mallet cursor offset math anchoring mallet contact point to mouse cursor.

- **Task Accomplished:** Engineered mouse cursor offset (mousePos.x - 42, mousePos.y - 6) aligning mallet strike head.

Prompt 3.24.22

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T20:34:46+07:00
- **Prompt Content:**

Implement lobbed projectile parabolic arc motion calculating 3D arc trajectory for pult plants.

- **Task Accomplished:** Engineered parabolic height displacement formula $y_{\text{arc}} = y_{\text{base}} - 4Ht^*(1-t)$ for lobbed shots.

Prompt 3.24.23

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T21:41:59+07:00
- **Prompt Content:**

Build Butter projectile impact handler applying 5-second complete freeze lock to hit zombie.

- **Task Accomplished:** Constructed Cornpult butter impact listener locking target zombie in butter block state.

Prompt 3.24.24

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T10:48:12+07:00
- **Prompt Content:**

Implement splash damage radius calculation dealing 1/3 damage to adjacent lane zombies.

- **Task Accomplished:** Engineered splash damage calculation inflicting partial damage to zombies within 1.5 tile radius.

Prompt 3.24.25

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T11:55:25+07:00
- **Prompt Content:**

Build particle system cleanup pool recycling inactive particle objects to prevent allocations.

- **Task Accomplished:** Created static ParticleEffect object pool recycling expired particle instances.

3.25 Memory Safety, Performance Optimization & Hot-Loop Refactoring

- **Conversation ID:** 908043dd-8ecd-4106-85ad-325bca0c
- **Category / Domain:** Performance Optimization & Refactoring
- **Primary Contributor:** Từ Hoàng Anh

- **Date Range (UTC+7):** 2026-08-29 to 2026-09-02
- **Total Active Prompts:** 20

Prompt 3.25.1

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T11:07:13+07:00
- **Prompt Content:**

Audit codebase hot loops to identify and eliminate per-frame heap allocations.

- **Task Accomplished:** Conducted memory profiling pass identifying dynamic string allocations inside 60 FPS render loops.

Prompt 3.25.2

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T12:14:26+07:00
- **Prompt Content:**

Refactor Resources::GetTexture to use stack char buffers and Small String Optimization (SSO).

- **Task Accomplished:** Optimized GetTexture string lookup path using char[128] stack buffers preventing dynamic malloc.

Prompt 3.25.3

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T13:21:39+07:00
- **Prompt Content:**

Replace dynamic std::string construction in BitmapFont kerning lookups with 16-bit integer maps.

- **Task Accomplished:** Replaced string pair keys with packed uint16_t integer keys in kerning dictionary lookups.

Prompt 3.25.4

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T14:28:52+07:00
- **Prompt Content:**

Implement lazy UI asset caching in MainMenu, OptionsMenu, and ShopMenu class headers.

- **Task Accomplished:** Cached Texture2D handles as mutable class members bypassing repetitive string resource lookups.

Prompt 3.25.5

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T15:35:05+07:00
- **Prompt Content:**

Refactor Reanimation track image resolution to use const std::string* pointers instead of copies.

- **Task Accomplished:** Eliminated per-frame string copies in Reanimation track rendering loops.

Prompt 3.25.6

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T16:42:18+07:00
- **Prompt Content:**

Fix keyframe array vector indexing with defensive clamping guarding against negative indices.

- **Task Accomplished:** Added std::max(0, index) clamping on all Reanim keyframe array index lookups.

Prompt 3.25.7

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-24T17:49:31+07:00
- **Prompt Content:**

Eliminate dynamic map allocation inside hot-loop geometry computation routines.

- **Task Accomplished:** Replaced dynamic map allocations with bounded static stack arrays in animation frame math.

Prompt 3.25.8

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T18:56:44+07:00
- **Prompt Content:**

Deduplicate per-frame track override map insertions in mouse hover event handlers.

- **Task Accomplished:** Added equality check before updating `m_trackImageOverrides` map preventing unnecessary rehashing.

Prompt 3.25.9

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T19:03:57+07:00
- **Prompt Content:**

Optimize Raylib `DrawTexturePro` rendering passes grouping identical texture quad draws.

- **Task Accomplished:** Batched sprite drawing calls to minimize GPU texture state switches.

Prompt 3.25.10

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T20:10:10+07:00
- **Prompt Content:**

Refactor entity vector iteration loops replacing index indexing with const reference iterators.

- **Task Accomplished:** Updated level update loops to use ranged-for const references reducing pointer indirection.

Prompt 3.25.11

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T21:17:23+07:00
- **Prompt Content:**

Fix potential memory leak in BitmapFont descriptor parser ensuring file streams close cleanly.

- **Task Accomplished:** Wrapped font file stream operations in RAII handles ensuring clean file closure.

Prompt 3.25.12

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T10:24:36+07:00
- **Prompt Content:**

Optimize collision detection spatial partition binning entity checks by lawn lane index.

- **Task Accomplished:** Gated projectile-zombie collision checks to entities sharing identical lane Y coordinates.

Prompt 3.25.13

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T11:31:49+07:00
- **Prompt Content:**

Refactor SunItem lifecycle management replacing dynamic allocations with object pooling.

- **Task Accomplished:** Implemented SunItem reuse pool recycling despawned sun instances.

Prompt 3.25.14

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-24T12:38:02+07:00
- **Prompt Content:**

Fix potential dangling pointer reference when removing dead plants from level entity vector.

- **Task Accomplished:** Updated plant cleanup pass using `std::remove_if` to safely prune destroyed plant objects.

Prompt 3.25.15

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-25T13:45:15+07:00
- **Prompt Content:**

Optimize wave spawner queue memory footprint pre-allocating wave zombie vector capacity.

- **Task Accomplished:** Pre-allocated vector memory for zombie wave queues preventing reallocations during waves.

Prompt 3.25.16

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-26T14:52:28+07:00
- **Prompt Content:**

Refactor Reanim XML line parsing replacing `std::string` substrings with `string_view` references.

- **Task Accomplished:** Optimized Reanim XML parser using C++17 `std::string_view` parsing primitives.

Prompt 3.25.17

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-27T15:59:41+07:00
- **Prompt Content:**

Add static assertion checks validating C++20 standard compliance and Raylib version compatibility.

- **Task Accomplished:** Added `static_assert` validation in main core headers checking compiler C++20 support.

Prompt 3.25.18

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-28T16:06:54+07:00
- **Prompt Content:**

Refactor UI button bounds calculation storing pre-computed `Rectangle` bounds in header fields.

- **Task Accomplished:** Pre-calculated UI button interaction rectangles during menu construction phase.

Prompt 3.25.19

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-29T17:13:07+07:00
- **Prompt Content:**

Fix audio stream buffer underrun by tuning audio frame buffer size in AudioManager.

- **Task Accomplished:** Adjusted Raylib InitAudioDevice buffer parameters ensuring glitch-free audio streaming.

Prompt 3.25.20

- **Member:** Từ Hoàng Anh
- **Sent At:** 2026-08-30T18:20:20+07:00
- **Prompt Content:**

Conduct comprehensive memory leak verification confirming zero memory leaks upon application exit.

- **Task Accomplished:** Verified application teardown sequence confirming all textures, fonts, and sounds unload cleanly.